

PSEUDOSPECTRAL TOPOLOGY OF COMPLEX TRIDIAGONAL TOEPLITZ MATRICES: THRESHOLDS AND FINITE-SIZE MONOTONICITY

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Abstract. For $T_n(\alpha, d, \beta) = \text{tridiag}(\alpha, d, \beta)$, with complex entries and $n \geq 2$, this paper classifies the connectedness of the strict Euclidean pseudospectrum and its dependence on the matrix order. Unitary diagonal conjugation, reversal, translation, rotation, and scaling reduce every nonscalar family to $A_n(a) = \text{tridiag}(a, 0, 1)$, where $a = \min\{|\alpha|, |\beta|\} / \max\{|\alpha|, |\beta|\}$. An abstract vertical-monotonicity principle first shows that a matrix with collinear spectrum whose least-singular-value sublevel sets contract toward the spectral line has contractible pseudospectral components, with connectedness governed by one least-singular-value barrier. For $0 < a < 1$, that barrier is $\max\{|\alpha|, |\beta|\}\gamma_n(a)$, where $\gamma_{n+1}(a) < \gamma_n(a)$ for every $n \geq 2$ and $\gamma_n(a) \asymp_a (n+1)a^{n/2}$. At the normal boundary $a = 1$, the threshold is an explicit half-gap and decreases strictly at the algebraic scale $\pi \max\{|\alpha|, |\beta|\} / (n+1)$. At the reducible boundary $a = 0$, the pseudospectrum is a disk and is connected at every order. Thus every component is contractible, connectedness persists with dimension, and the normal, nonnormal, and reducible regimes have distinct transition scales.

Key words. pseudospectrum, tridiagonal Toeplitz matrix, nonnormal matrix, least singular value, connectedness threshold, finite-size monotonicity

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1. Introduction. For $n \geq 1$, a matrix $A \in \mathbb{C}^{n \times n}$, and $\varepsilon > 0$, the strict Euclidean ε -pseudospectrum is

$$(1.1) \quad \sigma_\varepsilon(A) = \sigma(A) \cup \{z \in \mathbb{C} : \|(zI_n - A)^{-1}\|_2 > \varepsilon^{-1}\} = \{z \in \mathbb{C} : s_{\min}(zI_n - A) < \varepsilon\}.$$

Equivalently, it is the union of the spectra of $A + \Delta A$ over all complex perturbations satisfying $\|\Delta A\|_2 < \varepsilon$ [13]. Pseudospectral components therefore encode robust spectral clusters under an unstructured norm bound. Their merger is also tied to distance-to-defectivity questions. For matrices with distinct eigenvalues, first coalescence is characterized by a generalized saddle point of the smallest-singular-value function, and its level determines the spectral-norm distance to defectivity [1]. These observations motivate a finite-dimensional topological problem that is not determined by the spectrum alone.

The object of this paper is the full complex tridiagonal Toeplitz family

$$(1.2) \quad T_n(\alpha, d, \beta) = \text{tridiag}(\alpha, d, \beta) = \begin{bmatrix} d & \beta & & \\ \alpha & d & \ddots & \\ & \ddots & \ddots & \beta \\ & & \alpha & d \end{bmatrix}, \quad n \geq 2, \quad \alpha, d, \beta \in \mathbb{C},$$

where the arguments of tridiag are the subdiagonal, diagonal, and superdiagonal. Although the family has three complex parameters, its Euclidean pseudospectral topology has only two effective real parameters. When

$$c = \max\{|\alpha|, |\beta|\} > 0, \quad a = \frac{\min\{|\alpha|, |\beta|\}}{c} \in [0, 1],$$

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a diagonal unitary, possibly followed by reversal, reduces $T_n(\alpha, d, \beta)$ to a translate, rotation, and positive multiple of the canonical matrix

$$(1.3) \quad A_n(a) = \text{tridiag}(a, 0, 1).$$

Thus $a = 0$ is the reducible shift boundary, $0 < a < 1$ is the irreducible nonnormal regime, and $a = 1$ is the normal boundary. Treating all three cases gives a complete classification rather than a result for one selected orientation or choice of phases.

The finite-size difficulty is the comparison of global barriers defined on order-dependent spectral intervals. The diagonal similarity identifies the spectrum, while its condition number $a^{-(n-1)/2}$ leaves Euclidean least-singular-value geometry uncontrolled. Ordinary interlacing orders pointwise signed branches, whereas the barrier is a maximum over an order-dependent interval and its central gap changes parity. Toeplitz asymptotics identify the eventual scale rather than strict inequalities at adjacent finite orders. The argument below combines these tools with a strict comparison of the maximizing barrier at every consecutive order.

The first contribution is independent of Toeplitz structure. A general vertical-monotonicity theorem shows that, for any finite matrix with real spectrum, contraction of the least-singular-value level sets toward the real axis makes every pseudospectral component contractible and reduces connectedness to the largest backward-error barrier on the convex hull of the spectrum. Unitary-affine reduction transfers this statement to a matrix whose spectrum lies on any affine line.

For the nonnormal canonical regime $0 < a < 1$, write

$$\gamma_n(a) = \max_{|x| \leq 2\sqrt{a} \cos(\pi/(n+1))} s_{\min}(xI_n - A_n(a)).$$

The central finite-size result is $\gamma_{n+1}(a) < \gamma_n(a)$ for every $n \geq 2$. Explicit bounds give

$$(1.4) \quad \gamma_n(a) \asymp_a (n+1)a^{n/2},$$

including the sharp polynomial prefactor. At $a = 1$, the exact barrier is half the largest adjacent eigenvalue gap and decreases strictly at order $(n+1)^{-1}$. At $a = 0$, the pseudospectrum is a disk and is connected for every n . The classification therefore identifies a singular change from an exponential transition scale in the nonnormal interior to an algebraic scale at the normal boundary, followed by immediate connectedness at the reducible boundary.

The relevant literature has several complementary strands. A foundational spectral limit-set theorem for banded Toeplitz matrices is due to Schmidt and Spitzer [12]. Finite Toeplitz spectra, inverse norms, and pseudospectra are studied in [11, 5], and tridiagonal Toeplitz structure is treated in [10]. For general matrices, optimization methods for pseudospectra appear in [6], while component coalescence and nearest defective matrices are treated in [1]. Recent work on tridiagonal k -Toeplitz matrices addresses broader periodic coefficient patterns [3, 4]. Our contribution is a complete unitary-affine classification of complex tridiagonal Toeplitz pseudospectral connectedness, together with an exact finite-size order law in every irreducible regime and sharp transition asymptotics. The new finite-dimensional ingredient is the strict all- n barrier comparison, which complements the cited spectral and asymptotic results.

The proof separates general topology and unitary-affine geometry from the canonical nonnormal calculation. In the latter, a pentadiagonal determinant certificate establishes vertical monotonicity, and persymmetry turns singular values into signed eigenvalues of a symmetric pencil. Folded continuants, Chebyshev divided

differences, chord comparisons, and a separate central-gap certification prove strict finite-size comparison. Rectangular factorizations then give the sharp bounds and their logarithmic inversion. The longer algebraic certificates are isolated in [appendices A to F](#). A short physical correspondence appears briefly in [section 8](#).

2. A vertical-monotonicity principle. This section isolates the topology from the Toeplitz calculation. Let $n \geq 1$, let $B \in \mathbb{C}^{n \times n}$ have real spectrum, and put

$$h_B(x, y) = s_{\min}((x + iy)I_n - B), \quad x, y \in \mathbb{R}.$$

We say that B has the *vertical contraction property* if

$$(2.5) \quad h_B(x, ty) \leq h_B(x, y) \quad (x, y \in \mathbb{R}, 0 \leq t \leq 1).$$

This hypothesis is stated directly in the form needed by strict pseudospectral sublevel sets. For a real matrix it follows, for example, when $y \mapsto h_B(x, y)$ is nondecreasing on $[0, \infty)$, because singular values are invariant under complex conjugation.

Write

$$\lambda_-(B) = \min \sigma(B), \quad \lambda_+(B) = \max \sigma(B),$$

and define the real-axis barrier

$$(2.6) \quad \Gamma(B) = \max_{\lambda_-(B) \leq x \leq \lambda_+(B)} s_{\min}(xI_n - B).$$

The maximum exists by continuity. It is a global pointwise backward-error barrier: for each x ,

$$(2.7) \quad s_{\min}(xI_n - B) = \min\{\|\Delta B\|_2 : x \in \sigma(B + \Delta B)\}.$$

Indeed, if $(xI_n - B - \Delta B)v = 0$ for a unit vector v , then $s_{\min}(xI_n - B) \leq \|\Delta B\|_2$. Conversely, for unit singular vectors satisfying $(xI_n - B)v = s_{\min}(xI_n - B)u$, the rank-one choice $\Delta B = s_{\min}(xI_n - B)uv^*$ attains equality.

We use the following standard component fact in a form that applies to every finite matrix [\[6, 13\]](#).

LEMMA 2.1 (Eigenvalue in every component). *Let $n \geq 1$, $M \in \mathbb{C}^{n \times n}$, and $\varepsilon > 0$. Every connected component of $\sigma_\varepsilon(M)$ contains an eigenvalue of M .*

Proof. The estimate $s_{\min}(zI_n - M) \geq |z| - \|M\|_2$ makes the pseudospectrum bounded. Let $\mathcal{C}$ be a component containing no eigenvalue. If an eigenvalue belonged to $\bar{\mathcal{C}}$, continuity of the least singular value would give a pseudospectral ball meeting $\mathcal{C}$, and maximality would put that eigenvalue in $\mathcal{C}$. Thus $\bar{\mathcal{C}}$ lies in the resolvent set. On a neighborhood of this compact set, $z \mapsto \|(zI_n - M)^{-1}\|_2$ is continuous and subharmonic. It is at most ε^{-1} on $\partial\mathcal{C}$, whereas it is greater than ε^{-1} in $\mathcal{C}$. The maximum principle gives a contradiction. $\square$

THEOREM 2.2 (Vertical topology theorem). *Let $n \geq 1$. If $B \in \mathbb{C}^{n \times n}$ has real spectrum and satisfies [\(2.5\)](#), then, for every $\varepsilon > 0$, each connected component of $\sigma_\varepsilon(B)$ is contractible, and*

$$(2.8) \quad \sigma_\varepsilon(B) \text{ is connected} \iff [\lambda_-(B), \lambda_+(B)] \subset \sigma_\varepsilon(B) \iff \varepsilon > \Gamma(B).$$

If $\Gamma(B) > 0$, then at the positive level $\varepsilon = \Gamma(B)$, a maximizing real point is excluded from the strict pseudospectrum.

Proof. Put $\Omega = \sigma_\varepsilon(B)$. By (2.5),

$$\mathcal{H}_t(x + iy) = x + i(1 - t)y, \quad 0 \leq t \leq 1,$$

maps Ω into itself. For any component $\mathcal{C}$, the path $t \mapsto \mathcal{H}_t(z)$ cannot leave $\mathcal{C}$. Since $\mathcal{H}_t$ fixes the real axis, $\mathcal{H}_1(\mathcal{C}) = \mathcal{C} \cap \mathbb{R}$, and $\mathcal{H}_t|_{\mathcal{C}}$ is a strong deformation retraction onto this intersection. It is nonempty by Theorem 2.1; it is connected as the continuous image $\mathcal{H}_1(\mathcal{C})$, and is therefore an interval. Thus $\mathcal{C}$ is contractible.

If Ω is connected, then $\Omega \cap \mathbb{R}$ is an interval containing all eigenvalues and hence their convex hull. Conversely, if that convex hull lies in Ω , it belongs to one component and contains every eigenvalue. By Theorem 2.1, no other component can exist. This proves the first equivalence in (2.8); the second is exactly the strict sublevel definition and (2.6). $\square$

The theorem is invariant under a Euclidean similitude of the spectral plane: translation, rotation, and positive scaling merely translate, rotate, and scale the pseudospectrum and its barrier. This observation will allow the real-axis statement to govern every matrix in (1.2), whose spectrum lies on an affine line.

3. Complex tridiagonal Toeplitz matrices and main results. We begin with the transformations that remove the inessential phases, orientation, diagonal, and overall scale from (1.2). Let $V_n = \text{tridiag}(0, 0, 1)$, let J_n be the reversal permutation matrix, and, for $|q| = 1$, put

$$D_n(q) = \text{diag}(1, q, \dots, q^{n-1}).$$

PROPOSITION 3.1 (Unitary-affine reduction). *Let $n \geq 2$, $c = \max\{|\alpha|, |\beta|\} > 0$, and $a = \min\{|\alpha|, |\beta|\}/c$. There are a unitary matrix S_n and a unimodular number ω such that*

$$(3.9) \quad S_n^* T_n(\alpha, d, \beta) S_n = dI_n + c\omega A_n(a).$$

Consequently, for every $\varepsilon > 0$,

$$(3.10) \quad \sigma_\varepsilon(T_n(\alpha, d, \beta)) = d + c\omega \sigma_{\varepsilon/c}(A_n(a)).$$

Proof. First suppose $0 < |\alpha| \leq |\beta|$. Write $u_\alpha = \alpha/|\alpha|$ and $u_\beta = \beta/|\beta|$. Choose $|q| = 1$ with $q^2 = u_\alpha \overline{u_\beta}$, and put $\omega = u_\beta q = u_\alpha q^{-1}$. Direct multiplication gives

$$D_n(q)^* T_n(\alpha, d, \beta) D_n(q) = dI_n + |\beta|\omega A_n(|\alpha|/|\beta|).$$

If $|\alpha| > |\beta| > 0$, first conjugate by J_n , which interchanges the two off-diagonals, and apply the preceding case. If $\alpha = 0$ and $\beta \neq 0$, take $S_n = I_n$ and $\omega = \beta/|\beta|$; if $\beta = 0$ and $\alpha \neq 0$, take $S_n = J_n$ and $\omega = \alpha/|\alpha|$. Equation (3.10) follows because unitary conjugation preserves singular values, while

$$\sigma_\varepsilon(dI_n + c\omega A) = d + c\omega \sigma_{\varepsilon/c}(A). \quad \square$$

Thus the canonical parameter a covers both boundary cases. Their thresholds can be computed without the nonnormal argument. Define

$$(3.11) \quad \eta_n = \begin{cases} 2 \sin \frac{\pi}{2(n+1)}, & n \text{ even,} \\ \sin \frac{\pi}{n+1}, & n \text{ odd.} \end{cases}$$

PROPOSITION 3.2 (Reducible and normal boundaries). *Let $n \geq 2$ and $\varepsilon > 0$.*

1. *If $\alpha = \beta = 0$, then $\sigma_\varepsilon(T_n) = \{z : |z - d| < \varepsilon\}$. If exactly one off-diagonal is nonzero, then $\sigma_\varepsilon(T_n)$ is also an open disk centered at d . In both cases it is connected and contractible for every n ; its connectedness threshold is zero.*
2. *If $|\alpha| = |\beta| = c > 0$, then every pseudospectral component is contractible and*

$$(3.12) \quad \sigma_\varepsilon(T_n) \text{ is connected} \iff \varepsilon > c\eta_n.$$

Moreover, $\eta_{n+1} < \eta_n$ for every $n \geq 2$, and

$$(3.13) \quad 0 \leq \frac{\pi}{n+1} - \eta_n \leq \frac{\pi^3}{6(n+1)^3}.$$

In particular,

$$(3.14) \quad \eta_n = \frac{\pi}{n+1} + O(n^{-3}).$$

Proof. The scalar case is immediate. If exactly one off-diagonal is nonzero, [Theorem 3.1](#) reduces the matrix to the nilpotent Jordan shift V_n , up to a plane similitude. By [Theorem 2.1](#), every component of $\sigma_\varepsilon(V_n)$ contains its only eigenvalue, zero; hence the pseudospectrum is connected. Moreover, $D_n(q)^* V_n D_n(q) = qV_n$, while unitary invariance and $\sigma_\varepsilon(qV_n) = q\sigma_\varepsilon(V_n)$ show that it is invariant under every rotation. Its set of attained moduli is therefore a connected interval containing zero. Boundedness and openness make this interval $[0, R_{n,\varepsilon})$ for some $R_{n,\varepsilon} > 0$, and thus $\sigma_\varepsilon(V_n) = \{z : |z| < R_{n,\varepsilon}\}$. The plane similitude gives the asserted disk centered at d .

In part 2, [Theorem 3.1](#) reduces the problem to the real symmetric matrix $A_n(1)$, whose eigenvalues are $2\cos(k\pi/(n+1))$, $1 \leq k \leq n$. Its strict pseudospectrum is the union of open disks of radius ε/c about these collinear points. It is connected exactly when twice that radius exceeds every adjacent gap. The identity

$$\cos \frac{k\pi}{n+1} - \cos \frac{(k+1)\pi}{n+1} = 2 \sin \frac{(2k+1)\pi}{2(n+1)} \sin \frac{\pi}{2(n+1)}$$

shows that the varying sine factor has maximum 1 for even n and $\cos(\pi/(2(n+1)))$ for odd n . Hence half the largest gap is η_n in [\(3.11\)](#). At equality the two relevant open disks do not meet, so the inequality in [\(3.12\)](#) is strict. For this normal matrix,

$$s_{\min}((x + iy)I_n - A_n(1)) = \min_k \sqrt{(x - \lambda_k)^2 + y^2},$$

so the vertical contraction property is immediate. Component contractibility follows from [Theorem 2.2](#) and is preserved by the plane similitude in [\(3.10\)](#).

The two parity transitions are strict. From $2m$ to $2m+1$, if $x = \pi/(4m+2)$ and $y = \pi/(2m+2)$, then $y < 2x < \pi/2$, so $\sin y < \sin(2x) < 2\sin x$. From $2m+1$ to $2m+2$, put $p = m+1 \geq 2$, $x = \pi/(2p)$, and $y = \pi/(4p+2)$. The estimates $\sin x > x - x^3/6$, $\sin y < y$, and $24p^2 > \pi^2(2p+1)$ give $\sin x > 2\sin y$; the last inequality follows because $24p^2/(2p+1)$ is increasing for $p \geq 2$ and equals $96/5 > \pi^2$ at $p = 2$. Finally, $0 \leq x - \sin x \leq x^3/6$ for $x \geq 0$. Applied to the two parity formulas in [\(3.11\)](#), this gives [\(3.13\)](#), and hence [\(3.14\)](#). $\square$

We now state the nonnormal theorem in its canonical variables. Fix $0 < a < 1$, put $r = \sqrt{a}$, and write $A_n = A_n(a) = V_n + aV_n^T$. Define

$$\mathcal{G}_n(\varrho) = \text{diag}(1, \varrho, \dots, \varrho^{n-1}).$$

The diagonal similarity

$$(3.15) \quad A_n(a) = \mathcal{G}_n(r) r (V_n + V_n^T) \mathcal{G}_n(r)^{-1}$$

gives the displayed real spectrum, but is nonunitary for $0 < a < 1$, with $\kappa_2(\mathcal{G}_n(r)) = r^{-(n-1)}$. Define

$$I_n^{\text{sp}} = \left[-2r \cos \frac{\pi}{n+1}, 2r \cos \frac{\pi}{n+1} \right], \quad g_n(x) = s_{\min}(xI_n - A_n),$$

$$\gamma_n(a) = \max_{x \in I_n^{\text{sp}}} g_n(x), \quad N_c(a, \varepsilon) = \min\{n \geq 2 : \sigma_\varepsilon(A_n(a)) \text{ is connected}\}.$$

The theorem below shows that the minimum defining N_c exists. When a is fixed, its argument will be suppressed. The explicit lower and upper barriers are

$$(3.16) \quad \mathcal{U}_n(a) = r^n \csc \frac{\pi}{n+1},$$

$$(3.17) \quad \mathcal{L}_n(a) = \left(1 + r^2 - 2r \cos \frac{\pi}{n+1} \right) \frac{r^n(1-r^2)}{(1-r^{2(n+1)}) \sin \frac{3\pi}{2(n+1)}}.$$

THEOREM 3.3 (Canonical nonnormal theorem). *For every $0 < a < 1$ and $\varepsilon > 0$, the following assertions hold.*

1. *For every $n \geq 2$, every component of $\sigma_\varepsilon(A_n(a))$ is contractible.*
2. *For $n \geq 2$, the barrier sequence is strictly decreasing, $\gamma_{n+1}(a) < \gamma_n(a)$, tends to zero, and*

$$\sigma_\varepsilon(A_n(a)) \text{ is connected} \iff \varepsilon > \gamma_n(a).$$

3. *Connectedness persists after its first occurrence:*

$$\{n \geq 2 : \sigma_\varepsilon(A_n(a)) \text{ is connected}\} = \{n \geq N_c(a, \varepsilon)\}.$$

Moreover,

$$(3.18) \quad 1 + \max\left(\{1\} \cup \{n \geq 2 : \mathcal{L}_n(a) \geq \varepsilon\}\right) \leq N_c(a, \varepsilon),$$

$$N_c(a, \varepsilon) \leq \min\{n \geq 2 : \mathcal{U}_n(a) < \varepsilon\}.$$

4. *For fixed a , as $\varepsilon \downarrow 0$,*

$$(3.19) \quad N_c(a, \varepsilon) = \frac{2}{\log(1/a)} (\log(1/\varepsilon) + \log \log(1/\varepsilon)) + O_a(1),$$

and $N_c(a, \varepsilon) = 2$ if and only if $\varepsilon > a$.

Define the full-family threshold

$$(3.20) \quad \Theta_n(\alpha, \beta) = \begin{cases} 0, & \alpha\beta = 0, \\ c\gamma_n(a), & \alpha\beta \neq 0 \text{ and } |\alpha| \neq |\beta|, \\ c\eta_n, & |\alpha| = |\beta| = c > 0, \end{cases}$$

where the middle case uses $c = \max\{|\alpha|, |\beta|\}$ and $a = \min\{|\alpha|, |\beta|\}/c$. Also set

$$N_T(\alpha, \beta; \varepsilon) = \min\{n \geq 2 : \sigma_\varepsilon(T_n(\alpha, \beta)) \text{ is connected}\}.$$

The classification below shows that this minimum exists. By [Theorem 3.1](#), this value is independent of d and of the phases of the off-diagonal entries. Whenever $\alpha\beta \neq 0$, the notation below uses

$$c = \max\{|\alpha|, |\beta|\}, \quad a = \frac{\min\{|\alpha|, |\beta|\}}{c} \in (0, 1].$$

THEOREM 3.4 (Complex tridiagonal Toeplitz classification). *Let $\alpha, d, \beta \in \mathbb{C}$, $\varepsilon > 0$, and $n \geq 2$.*

1. *Every component of $\sigma_\varepsilon(T_n(\alpha, d, \beta))$ is contractible, and*

$$(3.21) \quad \sigma_\varepsilon(T_n(\alpha, d, \beta)) \text{ is connected} \iff \varepsilon > \Theta_n(\alpha, \beta).$$

2. *If $\alpha\beta \neq 0$, then $\Theta_{n+1}(\alpha, \beta) < \Theta_n(\alpha, \beta)$ and $\Theta_n(\alpha, \beta) \rightarrow 0$. Hence connectedness persists after its first occurrence. If $\alpha\beta = 0$, then the threshold is identically zero and the pseudospectrum is already connected at every order, so persistence is immediate.*

3. *In the irreducible nonnormal case $0 < a < 1$,*

$$(3.22) \quad N_T(\alpha, \beta; \varepsilon) = N_c(a, \varepsilon/c) = \frac{2}{\log(1/a)} \left(\log \frac{c}{\varepsilon} + \log \log \frac{c}{\varepsilon} \right) + O_a(1)$$

as $\varepsilon \downarrow 0$. In the normal case $a = 1$,

$$(3.23) \quad N_T(\alpha, \beta; \varepsilon) = \frac{\pi c}{\varepsilon} + O(1).$$

If $\alpha\beta = 0$, then $N_T(\alpha, \beta; \varepsilon) = 2$.

Proof. For $c > 0$, apply [Theorem 3.1](#). The nonnormal assertions follow from [Theorem 3.3](#), with ε replaced by ε/c ; the boundary assertions follow from [Theorem 3.2](#). In the normal case, put $m = N_T(\alpha, \beta; \varepsilon)$. For sufficiently small ε , one has $m > 2$ and

$$c\eta_{m-1} \geq \varepsilon > c\eta_m.$$

The bound [\(3.13\)](#) first gives $m \sim \pi c/\varepsilon$, and then $\varepsilon = \pi c/m + O(m^{-2})$. Hence $m = \pi c/\varepsilon + O(1)$, proving [\(3.23\)](#). If $c = 0$, the scalar case applies directly. $\square$

4. Vertical monotonicity and topological reduction. The topological reduction rests on a monotonicity property that makes every pseudospectral component connection visible on the real axis.

LEMMA 4.1 (Vertical monotonicity). *Let $n \geq 2$ and $0 < a < 1$. For every real x ,*

$$y \mapsto s_{\min}((x + iy)I_n - A_n), \quad y \geq 0,$$

is nondecreasing.

Proof. Put $Y = y^2$, and, for $t \geq 0$, define

$$\begin{aligned} \mathcal{P}_n(Y, t) &= ((x + i\sqrt{Y})I_n - A_n)^*((x + i\sqrt{Y})I_n - A_n) - t^2 I_n, \\ F_n(Y, t) &= \det \mathcal{P}_n(Y, t). \end{aligned}$$

[Theorem A.1](#) proves the finite-dimensional implication

$$(4.24) \quad \mathcal{P}_n(Y, t) > 0 \implies \partial_Y F_n(Y, t) > 0.$$

Finally fix $Y_0 \geq 0$. When the least singular value at Y_0 equals zero, monotonicity follows from nonnegativity of singular values. Otherwise take $0 \leq t < s_{\min}((x + i\sqrt{Y_0})I_n - A_n)$. On the maximal interval to the right of

Y_0 on which $\mathcal{P}_n(Y, t) > 0$, implication (4.24) makes its determinant strictly increasing. At any finite right endpoint, continuity of Hermitian eigenvalues would produce a positive-semidefinite singular matrix with determinant zero. Strict increase from the positive determinant at Y_0 precludes such an endpoint. Thus $\mathcal{P}_n(Y, t) > 0$ for every $Y \geq Y_0$. Letting t increase to the least singular value at Y_0 proves the lemma. $\square$

Because $A_n(a)$ is real, Theorem 4.1 gives the vertical contraction property (2.5). Its spectrum is the real set $\{2\sqrt{a} \cos(k\pi/(n+1)) : 1 \leq k \leq n\}$, whose convex hull is I_n^{sp} . Applying Theorem 2.2 therefore gives contractibility of every component and

$$(4.25) \quad \sigma_\varepsilon(A_n(a)) \text{ is connected} \iff I_n^{\text{sp}} \subset \sigma_\varepsilon(A_n(a)) \iff \gamma_n(a) < \varepsilon.$$

The strict inequality is essential: at $\varepsilon = \gamma_n(a)$, a maximizing real point remains excluded.

5. Strict finite-size comparison of real-gap barriers. The final barrier can occur in any inter-eigenvalue gap, and the identity of the central gap changes with parity. A global finite-size comparison must therefore control every gap. Persymmetry turns singular values into signed eigenvalue branches, folded Chebyshev coordinates parametrize those branches, scale inequalities compare noncentral gaps, and a separate interlacing argument handles the parity-changing central gaps. We first define the local real-gap barriers. Write the eigenvalues in decreasing order as

$$\lambda_{n,k} = 2r \cos \frac{k\pi}{n+1}, \quad 1 \leq k \leq n,$$

and call

$$G_{n,k} = [\lambda_{n,k+1}, \lambda_{n,k}], \quad \gamma_{n,k} = \max_{x \in G_{n,k}} g_n(x), \quad 1 \leq k < n,$$

the closed spectral gaps and their heights. Continuity guarantees that the maximum on each compact gap is attained, while the spectral endpoints have height zero. Every interior gap point belongs to the resolvent set, so $xI_n - A_n$ is invertible there and $g_n(x) > 0$. Consequently every gap height is positive and every maximizer lies in the open gap. Hence

$$(5.26) \quad \gamma_n = \max_{1 \leq k < n} \gamma_{n,k}.$$

LEMMA 5.1 (Strict barrier comparison). *For every $n \geq 2$,*

$$\gamma_{n+1} < \gamma_n.$$

Let

$$J_n = [\mathbf{1}_{\{i+j=n+1\}}]_{i,j=1}^n$$

be the reversal matrix. Persymmetry supplies the signed spectral representation needed for the finite-size comparison: $A_n J_n = J_n A_n^T$, so $B_n(x) = (xI_n - A_n)J_n$ is real symmetric. Moreover,

$$(xI_n - A_n - sJ_n)J_n = B_n(x) - sI_n.$$

Since J_n is orthogonal, the absolute eigenvalues of $B_n(x)$ are exactly the singular values of $xI_n - A_n$, including multiplicity. Consequently,

$$(5.27) \quad s_{\min}(xI_n - A_n) = \min\{|s| : \det(xI_n - A_n - sJ_n) = 0\}.$$

Let $q_n(x, s) = \det(xI_n - A_n - sJ_n)$.

LEMMA 5.2 (The middle signed branch). *Write the eigenvalues of $B_n(x)$ in decreasing order as $\alpha_1(x) \geq \dots \geq \alpha_n(x)$, and abbreviate $\lambda_k = \lambda_{n,k}$. If $\lambda_{j+1} < x < \lambda_j$, then*

$$s_{\min}(xI_n - A_n) = |s_*(x)|, \quad \text{sgn}(s_*(x)) = (-1)^j,$$

where

$$s_*(x) = \begin{cases} \alpha_m(x), & n = 2m, \\ \alpha_{m+1}(x), & n = 2m + 1. \end{cases}$$

Proof. Let $A_n^{\text{sym}} = r(V_n + V_n^T)$. The identities

$$\begin{aligned} A_n &= \mathcal{G}_n(r) A_n^{\text{sym}} \mathcal{G}_n(r)^{-1}, \\ \mathcal{G}_n(r)^{-1} J_n &= r^{-(n-1)} J_n \mathcal{G}_n(r) \end{aligned}$$

give

$$(5.28) \quad B_n(x) = r^{-(n-1)} \mathcal{G}_n(r) (xI_n - A_n^{\text{sym}}) J_n \mathcal{G}_n(r).$$

The sine eigenvector of A_n^{sym} belonging to λ_k is an eigenvector of J_n with eigenvalue $(-1)^{k+1}$. Hence (5.28) and Sylvester's law of inertia show that $B_n(x)$ has the same inertia as the diagonal matrix with entries

$$(-1)^{k+1}(x - \lambda_k), \quad 1 \leq k \leq n.$$

Counting signs gives

$$(5.29) \quad \nu_+(B_{2m}(x)) = \begin{cases} m, & j \text{ even}, \\ m-1, & j \text{ odd}, \end{cases} \quad \nu_+(B_{2m+1}(x)) = \begin{cases} m+1, & j \text{ even}, \\ m, & j \text{ odd}. \end{cases}$$

First let $n = 2m$. Permute the coordinates into the two groups

$$(1, \dots, m), \quad (2m, 2m-1, \dots, m+1).$$

Here $\mathbf{e}_j$ denotes the j th Euclidean coordinate vector in the ambient space. In these coordinates

$$(5.30) \quad \begin{aligned} B_{2m}(x) &= \hat{P}_m(x) - \mathcal{R}_{\text{bd}}, \\ \hat{P}_m(x) &= \begin{bmatrix} 0 & P_m(x) \\ P_m(x)^T & 0 \end{bmatrix}, \\ \mathcal{R}_{\text{bd}} &= \begin{bmatrix} \mathbf{e}_m \mathbf{e}_m^T & 0 \\ 0 & a \mathbf{e}_m \mathbf{e}_m^T \end{bmatrix}, \end{aligned}$$

where $P_m(x) = xI_m - A_m$. Thus $\mathcal{R}_{\text{bd}} \geq 0$ and $\text{rank } \mathcal{R}_{\text{bd}} = 2$. For $\Gamma = \text{diag}(I_m, -I_m)$,

$$\tilde{B} = -\Gamma B_{2m}(x) \Gamma = \hat{P}_m(x) + \mathcal{R}_{\text{bd}} = B_{2m}(x) + 2\mathcal{R}_{\text{bd}}.$$

If $\tilde{\alpha}_1 \geq \dots \geq \tilde{\alpha}_{2m}$ are the eigenvalues of $\tilde{B}$, then

$$(5.31) \quad \tilde{\alpha}_k = -\alpha_{2m+1-k}.$$

Positive-semidefinite rank-two interlacing gives

$$(5.32) \quad \tilde{\alpha}_k \geq \alpha_k \geq \tilde{\alpha}_{k+2}, \quad 1 \leq k \leq 2m-2.$$

If j is even, the only candidates for minimum modulus have moduli α_m and $-\alpha_{m+1} = \tilde{\alpha}_m$, and $\alpha_m \leq \tilde{\alpha}_m$. If j is odd and $m \geq 2$, the two candidates have moduli α_{m-1} and $-\alpha_m = \tilde{\alpha}_{m+1}$; taking $k = m - 1$ in (5.32) gives $\alpha_{m-1} \geq \tilde{\alpha}_{m+1}$. For $m = 1$, both eigenvalues in the sole gap are negative, so the larger one α_1 has minimum modulus. Thus α_m always attains the minimum modulus, and its sign follows from (5.29). Equality in an interlacing comparison is harmless: it only gives a multiple least singular value.

Now let $n = 2m + 1$. Group the first and last m coordinates as above and leave the middle coordinate last. Then

$$(5.33) \quad B_{2m+1}(x) = \begin{bmatrix} \hat{P}_m(x) & \mathbf{w} \\ \mathbf{w}^T & x \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} -\mathbf{e}_m \\ -a\mathbf{e}_m \end{bmatrix}.$$

In an open spectral gap of A_{2m+1} , $P_m(x)$ is nonsingular. Indeed, the explicit path spectrum gives

$$\text{spec}(A_m) = \left\{ 2r \cos \frac{\ell\pi}{m+1} : 1 \leq \ell \leq m \right\} = \{ \lambda_{2m+1, 2\ell} : 1 \leq \ell \leq m \},$$

so its zeros are exactly the even-indexed eigenvalues of A_{2m+1} , none of which lies in an open spectral gap. The eigenvalues of the principal submatrix $\hat{P}_m(x)$ are consequently

$$s_1(P_m) \geq \cdots \geq s_m(P_m) > 0 > -s_m(P_m) \geq \cdots \geq -s_1(P_m).$$

Cauchy interlacing gives

$$\alpha_m \geq s_m(P_m) \geq \alpha_{m+1} \geq -s_m(P_m) \geq \alpha_{m+2}.$$

If j is even, (5.29) and interlacing give

$$0 < \alpha_{m+1} \leq s_m(P_m) \leq -\alpha_{m+2};$$

if j is odd, they give

$$0 < -\alpha_{m+1} \leq s_m(P_m) \leq \alpha_m.$$

Every other eigenvalue is farther from zero. Hence α_{m+1} attains the minimum modulus, with sign prescribed by (5.29); equality again only means multiplicity. At a spectral endpoint the zero is simple because A_n is similar to the Jacobi matrix A_n^{sym} , whose eigenvalues are simple. Thus the selected ordered branch extends continuously to zero at both ends of the closed gap. This proves the assertion. $\square$

We use folded coordinates for the noncentral gaps and a separate interlacing argument for the parity-changing central gaps.

Folded continuants

Let U_j be the Chebyshev polynomial of the second kind, with $U_{-1} = 0$. Put

$$C_0 = x^2 + (1-a)^2 - s^2, \quad \zeta_s = \frac{1 + a^2 + (1+a)s}{2a},$$

and define the folded discriminant and recurrence coefficient by

$$\Delta_{\text{fold}} = ((1+a-x)^2 - s^2)((1+a+x)^2 - s^2),$$

$$\mathfrak{b} = (1+a^2)x^2 + 2as^2 + 2a^2 - 2a - 2a^3.$$

The global recurrence polynomial is

$$\begin{aligned}
(5.34) \quad \Pi(z) &:= 1 - C_0 z + \mathfrak{b} z^2 - a^2 C_0 z^3 + a^4 z^4, \\
u, v \in \mathbb{R}, \quad 2a(u+v) &= C_0, \quad a^2(4uv+2) = \mathfrak{b}, \\
\implies \Pi(z) &= (1 - 2auz + a^2 z^2)(1 - 2avz + a^2 z^2).
\end{aligned}$$

On a real sheet with $\Delta_{\text{fold}} \geq 0$, we additionally put

$$\xi_{\pm} = \frac{C_0 \pm \sqrt{\Delta_{\text{fold}}}}{4a}.$$

Then the pair $(u, v) = (\xi_+, \xi_-)$ satisfies the two symmetric relations in (5.34):

$$\xi_+ + \xi_- = \frac{C_0}{2a}, \quad \mathfrak{b} = a^2(4\xi_+\xi_- + 2).$$

We use the root presentation on these real sheets and the polynomial Π itself for the global transfer and generating identities. The complete five-minor transfer, characteristic-polynomial calculation, boundary residuals, and generating functions are given in [appendix B](#). Their parity-unified conclusion is the part needed in the main argument. For a differentiable real function f , write

$$[f]_{u,v} := \begin{cases} \frac{f(u) - f(v)}{u - v}, & u \neq v, \\ f'(u), & u = v \end{cases}.$$

Whenever the real pair u, v satisfies the two symmetric relations in (5.34), the divided-difference calculation in [appendix C](#) gives

$$(5.35) \quad q_{2m}(x, s) = a^m [\mathfrak{f}_m]_{u,v}, \quad \mathfrak{f}_m(t) = (t - \zeta_s)(U_m(t) + U_{m-1}(t)),$$

and

$$(5.36) \quad q_{2m+1}(x, s) = a^m [\mathfrak{g}_m]_{u,v}, \quad \mathfrak{g}_m(t) = \left((x - s)t - \frac{x(1 + a^2) + 2as}{2a} \right) U_m(t).$$

The recurrence and generating identities are polynomial and therefore global. The real-pair formulas are used only after the two symmetric relations have been checked; at $u = v$, the displayed derivative is the collision convention. For $\Delta_{\text{fold}} < 0$, the argument stays at the polynomial level and avoids any complex square-root branch.

These formulas replace the singular-value problem by a signed scalar branch with parity already resolved. The next step identifies a geometric parametrization of that branch.

Half-angle chord parametrization

Put $h = -\frac{1}{2} \log a > 0$, so that $a = e^{-2h}$, and set

$$\mathfrak{s}_h = \sinh h, \quad \mathfrak{c}_h = \cosh h.$$

For $K \geq 2$ and $0 \leq \theta \leq \pi$, define

$$(5.37) \quad \Phi_K(\theta) = \sqrt{\mathfrak{s}_h^2 + \sin^2 \theta} \frac{|\sin(K\theta)|}{\sin \theta},$$

where the quotient is interpreted continuously at $\theta = 0, \pi$, and

$$\Psi_K(\eta) = \sqrt{\mathfrak{s}_h^2 - \sinh^2 \eta} \frac{\sinh(K\eta)}{\sinh \eta}, \quad 0 \leq \eta \leq h,$$

with the limiting value at $\eta = 0$. The following half-angle reduction also fixes the sign convention. Put

$$y = \cos \theta, \quad z = \cosh \eta, \quad \mathfrak{p}_\theta = \sqrt{\mathfrak{c}_h^2 - y^2}, \quad \mathfrak{p}_\eta = \sqrt{\mathfrak{c}_h^2 - z^2},$$

and let $L_0 = 2r/\mathfrak{c}_h$. If $x = L_0 y z$, $s = L_0 \mathfrak{p}_\theta \mathfrak{p}_\eta > 0$, and $\tau = \varsigma s$ with $\varsigma \in \{-1, 1\}$, then direct expansion gives

$$(5.38) \quad \begin{aligned} (1 + a + x)^2 - s^2 &= 4a(z + y)^2, & (1 + a - x)^2 - s^2 &= 4a(z - y)^2, \\ u = 2z^2 - 1, & v = 2y^2 - 1, & \zeta_\tau &= 2\mathfrak{c}_h^2 - 1 + 2\varsigma \mathfrak{p}_\theta \mathfrak{p}_\eta. \end{aligned}$$

The real pair (u, v) satisfies the two symmetric relations in (5.34). For $K = 2m + 1$, use

$$U_m(2t^2 - 1) + U_{m-1}(2t^2 - 1) = U_{K-1}(t);$$

for $K = 2m + 2$, use

$$U_m(2t^2 - 1) = \frac{U_{K-1}(t)}{2t}.$$

Substitution of this pair in (5.35) and (5.36), respectively, reduces either equation $q_{K-1}(x, \tau) = 0$ to

$$(5.39) \quad \mathfrak{p}_\theta U_{K-1}(y) = \varsigma \mathfrak{p}_\eta U_{K-1}(z).$$

In the even-fold case the intermediate factorization is

$$(\mathfrak{p}_\eta + \varsigma \mathfrak{p}_\theta)(\mathfrak{p}_\eta U_{K-1}(z) - \varsigma \mathfrak{p}_\theta U_{K-1}(y)) = 0.$$

The first factor could vanish only if $\varsigma = -1$ and $\mathfrak{p}_\eta = \mathfrak{p}_\theta$, which would imply $y^2 = z^2$; the distinct-variable condition therefore leaves precisely (5.39). In the odd-fold case the reduction also contains the factor $\mathfrak{p}_\eta y + \varsigma \mathfrak{p}_\theta z$. On a distinct-variable sheet this factor is nonzero: if it vanished, squaring would give

$$(\mathfrak{c}_h^2 - z^2)y^2 = (\mathfrak{c}_h^2 - y^2)z^2,$$

and hence $y^2 = z^2$. Thus the factor is nonzero whenever the folded variables are distinct. On the outer lobe $U_{K-1}(z) > 0$, whereas, for $j\pi/K < \theta < (j+1)\pi/K$,

$$\operatorname{sgn}(U_{K-1}(\cos \theta)) = \operatorname{sgn}(\sin(K\theta)) = (-1)^j.$$

Moreover, $\mathfrak{p}_\theta |U_{K-1}(y)| = \Phi_K(\theta)$ and $\mathfrak{p}_\eta U_{K-1}(z) = \Psi_K(\eta)$. Thus (5.39) is equivalent to $\varsigma = (-1)^j$ and

$$(5.40) \quad \Phi_K(\theta) = \Psi_K(\eta),$$

with

$$(5.41) \quad s = \mathcal{S}_{\text{hyp}}(\theta, \eta) := \frac{2r}{\mathfrak{c}_h} \sqrt{(\mathfrak{s}_h^2 + \sin^2 \theta)(\mathfrak{s}_h^2 - \sinh^2 \eta)}, \quad x = \frac{2r}{\mathfrak{c}_h} \cos \theta \cosh \eta.$$

Consequently $q_{K-1}(x, (-1)^j s) = 0$. Conversely, for distinct real folded variables, their sum and product determine

$$x^2 = L_0^2 y^2 z^2, \quad s^2 = L_0^2 (\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2).$$

Thus, for distinct real variables and a fixed sign of x , the preceding steps are reversible, and the two chord relations are equivalent to the corresponding signed equation. The determinant identities themselves extend to a folded-variable collision by the continuous divided-difference convention, but we invoke the chord equivalence only for distinct variables; the selected middle branch is proved below to be collision-free. Here $K - 1$ is the matrix size. If the exterior argument has crossed through 1, write it as $z = \cos \phi$, $0 \leq \phi < \pi/K$. Then the chord equation is $\Phi_K(\theta) = \Phi_K(\phi)$, and the second factor in the formula for s is $\mathfrak{s}_h^2 + \sin^2 \phi$. We denote the resulting value by

$$\mathcal{S}_{\text{ell}}(\theta, \phi) = \frac{2r}{\mathfrak{c}_h} \sqrt{(\mathfrak{s}_h^2 + \sin^2 \theta)(\mathfrak{s}_h^2 + \sin^2 \phi)}.$$

Branch classification and outer-lobe geometry

It remains to identify which chords realize the middle branch of [Theorem 5.2](#). Set

$$(5.42) \quad \mathcal{W}_K(u) = \sqrt{\mathfrak{c}_h^2 - u^2} |U_{K-1}(u)|, \quad |u| \leq \mathfrak{c}_h.$$

At $u = \pm \mathfrak{c}_h$, this denotes the continuous endpoint value 0. Then

$$\Phi_K(\theta) = \mathcal{W}_K(\cos \theta), \quad \Psi_K(\eta) = \mathcal{W}_K(\cosh \eta).$$

If $u_{\ell,K}^{(0)} = \cos \frac{\ell\pi}{K}$, then, away from the zeros,

$$\frac{d}{du} \log \mathcal{W}_K(u) = \sum_{\ell=1}^{K-1} \frac{1}{u - u_{\ell,K}^{(0)}} - \frac{u}{\mathfrak{c}_h^2 - u^2},$$

and its derivative is

$$(5.43) \quad - \sum_{\ell=1}^{K-1} \frac{1}{(u - u_{\ell,K}^{(0)})^2} - \frac{\mathfrak{c}_h^2 + u^2}{(\mathfrak{c}_h^2 - u^2)^2} < 0.$$

At each finite nodal endpoint the logarithmic derivative tends to the appropriate signed infinity, and it tends to $-\infty$ as $u \uparrow \mathfrak{c}_h$. Thus there is exactly one lobe maximum in every nodal interval, including the outer interval $\mathcal{J}_K^{\text{out}} = (\cos \frac{\pi}{K}, \mathfrak{c}_h)$. Let u_K^{max} be the unique maximizer on $\mathcal{J}_K^{\text{out}}$. For every $0 \leq \ell < \mathcal{W}_K(u_K^{\text{max}})$, there is a unique solution

$$z_K(\ell) \in (u_K^{\text{max}}, \mathfrak{c}_h], \quad \mathcal{W}_K(z_K(\ell)) = \ell;$$

we call it the *endpoint-side outer solution*. If $z_K(\ell) \geq 1$, write it as $\cosh \eta$, $0 \leq \eta \leq h$; if it is at most 1, write it as $\cos \phi$, $0 \leq \phi < \pi/K$. These are two coordinates on the same outer lobe, joined at $z = 1$.

For a positive-half gap and $\theta \in (j\pi/K, (j+1)\pi/K)$, $\theta \leq \pi/2$, a j -chord is the pair

$$y = \cos \theta, \quad z = z_K(\mathcal{W}_K(y)), \quad \mathcal{W}_K(y) = \mathcal{W}_K(z),$$

together with the coordinate-free values

$$x = L_0 y z, \quad s = L_0 \sqrt{(\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2)};$$

its signed root is $(-1)^j s$. The term endpoint-side always refers to the part of the outer lobe adjacent to $\mathfrak{c}_h$, independently of whether it is described by the hyperbolic or elliptic coordinate.

Reflection reduces the branch classification to a positive-half gap, so $j \geq 1$ and $\theta \leq \pi/2$. Put $\phi_{\text{rem}} = \theta - j\pi/K \in (0, \pi/K)$. Since $|\sin(K\phi_{\text{rem}})| = |\sin(K\theta)|$ and

$$\mathcal{A}_h(t) = \frac{\sqrt{\mathfrak{s}_h^2 + \sin^2 t}}{\sin t}$$

is strictly decreasing on $(0, \pi/2]$,

$$(5.44) \quad \Phi_K(\theta) < \Phi_K(\phi_{\text{rem}}) \leq \max_{u \in \mathcal{J}_{K^{\text{out}}}^{\text{out}}} \mathcal{W}_K(u).$$

Thus the endpoint-side outer solution exists for every inner parameter.

At $x = \lambda_{K-1,k} = 2r \cos \frac{k\pi}{K}$, $s = 0$, the two folded variables are $\cos \frac{2k\pi}{K}$ and $\cosh(2h)$, or, in half variables, $\cos \frac{k\pi}{K}$ and $\mathfrak{c}_h$. Moreover, zero is a simple eigenvalue of $B_{K-1}(\lambda_{K-1,k})$: the matrix $\lambda_{K-1,k}I_{K-1} - A_{K-1}$ has rank $K-2$, and J_{K-1} is invertible. The implicit-function theorem therefore gives, near either gap endpoint, a real continuation of the middle signed root and real half variables $y = \cos \theta$ and z .

Chord-sheet exhaustion

By [Theorem D.1](#), the ordered middle branch in every positive gap is exactly the selected endpoint-side chord sheet. In particular, its signed root is $(-1)^j s$, its magnitude is the least singular value, and its folded variables are distinct in the open gap. The same result covers the positive half of the central gap when K is odd.

LEMMA 5.3 (Noncentral gap comparison). *Let $j \geq 1$ and $N \geq 2(j+1)$ be integers, and put $M = N+1$. For every $x \in G_{M-1,j}^{\circ}$, there is $\hat{x} \in G_{N-1,j}^{\circ}$ such that*

$$g_{N-1}(\hat{x}) > g_{M-1}(x).$$

Proof. Fix $x \in G_{M-1,j}^{\circ}$. By [Theorem D.1](#), its middle branch has a hyperbolic or elliptic chord representation with

$$\theta \in \left(\frac{j\pi}{M}, \frac{(j+1)\pi}{M} \right).$$

Set $\kappa = N/M$ and $\theta_0 = \theta/\kappa$. The hypotheses place θ_0 in the corresponding N -nodal interval, with every trigonometric argument below in $(0, \pi/2]$.

First suppose $\Phi_M(\theta) = \Psi_M(\eta)$. Concavity of sine gives

$$(5.45) \quad \frac{\Phi_N(\theta_0)}{\Phi_M(\theta)} = \frac{\mathcal{A}_h(\theta_0)}{\mathcal{A}_h(\theta)} > \kappa.$$

Indeed, $\theta = \kappa\theta_0$ and strict concavity gives $\sin \theta > \kappa \sin \theta_0$; hence

$$\mathcal{A}_h(\theta_0)^2 - \kappa^2 \mathcal{A}_h(\theta)^2 = (1 - \kappa^2) + \mathfrak{s}_h^2 \left(\frac{1}{\sin^2 \theta_0} - \frac{\kappa^2}{\sin^2 \theta} \right) > 0.$$

On the other hand,

$$(5.46) \quad \frac{\Psi_N(\eta)}{\Psi_M(\eta)} = \frac{\sinh(N\eta)}{\sinh(M\eta)} \leq \kappa,$$

with strict inequality for $\eta > 0$, by convexity of $\sinh$ on the positive half-axis. Hence $\Phi_N(\theta_0) > \Psi_N(\eta)$. Since Φ_N vanishes at the right endpoint $(j+1)\pi/N$, its rightmost solution $\widehat{\theta}$ of

$$\Phi_N(\widehat{\theta}) = \Psi_N(\eta)$$

exists by the intermediate value theorem and satisfies $\widehat{\theta} > \theta_0 > \theta$. Strict log-concavity of $\mathcal{W}_N$ makes it the unique solution on the descending half of that lobe. The same exterior variable is endpoint-side for the N -problem as well. Indeed, for $u = \cosh \eta > 1$,

$$\frac{d}{du} \log \mathcal{W}_M(u) - \frac{d}{du} \log \mathcal{W}_N(u) = \frac{M \coth(M\eta) - N \coth(N\eta)}{\sinh \eta} > 0.$$

The numerator is positive because, for $\eta > 0$,

$$\frac{d}{dt} (t \coth(t\eta)) = \frac{\sinh(t\eta) \cosh(t\eta) - t\eta}{\sinh^2(t\eta)} > 0.$$

The endpoint-side branch is characterized by a nonpositive logarithmic derivative. The displayed difference therefore makes the N -derivative strictly negative. The case $\eta = 0$ follows by continuity. The chord classification above says that the constructed value $\mathcal{S}_{\text{hyp}}(\widehat{\theta}, \eta)$ is the least singular value at its constructed abscissa. Formula (5.41) and the strict increase of sine on the interval now give

$$(5.47) \quad \mathcal{S}_{\text{hyp}}(\widehat{\theta}, \eta) > \mathcal{S}_{\text{hyp}}(\theta, \eta).$$

Suppose next that the exterior endpoint is elliptic, so that $\Phi_M(\theta) = \Phi_M(\phi)$, $0 < \phi < \pi/M$. (The case $\phi = 0$ is the limiting hyperbolic case already covered.) Put

$$\theta_0 = \theta/\kappa, \quad \phi_0 = \phi/\kappa.$$

The ratio

$$\mathcal{A}_{N,M}^\Phi(t) = \frac{\Phi_N(t/\kappa)}{\Phi_M(t)} = \frac{\mathcal{A}_h(t/\kappa)}{\mathcal{A}_h(t)}$$

is strictly increasing while $0 < t < t/\kappa \leq \pi/2$. Indeed,

$$\frac{d}{dt} \log \mathcal{A}_{N,M}^\Phi(t) = \mathfrak{d}_h(t) - \kappa^{-1} \mathfrak{d}_h(t/\kappa) > 0, \quad \mathfrak{d}_h(t) = \frac{\mathfrak{s}_h^2 \cot t}{\mathfrak{s}_h^2 + \sin^2 t},$$

because

$$t \mathfrak{d}_h(t) = (t \cot t) \frac{\mathfrak{s}_h^2}{\mathfrak{s}_h^2 + \sin^2 t}$$

is strictly decreasing: both positive factors decrease on $(0, \pi/2)$, and the second decreases strictly. Since $\theta > \pi/M > \phi$, it follows that $\Phi_N(\theta_0) > \Phi_N(\phi_0)$. The rightmost solution of

$$\Phi_N(\widehat{\theta}) = \Phi_N(\phi_0)$$

therefore has $\widehat{\theta} > \theta_0$. Its exterior variable is endpoint-side. Indeed, if $\Lambda_K(t) = (\log \Phi_K)'(t)$, then on the endpoint-side part of the first outer lobe $\Lambda_M(\phi) \geq 0$, and differentiation of $\Phi_N(t/\kappa) = \mathcal{A}_{N,M}^\Phi(t) \Phi_M(t)$ gives

$$\kappa^{-1} \Lambda_N(t/\kappa) = \Lambda_M(t) + \frac{(\mathcal{A}_{N,M}^\Phi)'(t)}{\mathcal{A}_{N,M}^\Phi(t)} > 0.$$

Because $\Lambda_K(\phi) = -\sin \phi (\log \mathcal{W}_K)'(\cos \phi)$, positivity of $\Lambda_N(\phi_0)$ is exactly the endpoint-side condition for the N -outer solution. The one-maximum property now fixes the label, and the chord classification says again that $\mathcal{S}_{\text{ell}}(\hat{\theta}, \phi_0)$ is the least singular value at the constructed abscissa. Here $\hat{\theta} > \theta_0 > \theta$ and $\phi_0 > \phi$, with all four angles below $\pi/2$. Thus both sine factors in the elliptic version of (5.41) have increased, at least one strictly, and hence

$$(5.48) \quad \mathcal{S}_{\text{ell}}(\hat{\theta}, \phi_0) > \mathcal{S}_{\text{ell}}(\theta, \phi).$$

Equations (5.47) and (5.48), together with the coordinate formula for x , place the constructed abscissa $\hat{x}$ in $G_{N-1,j}^\circ$. Chord exhaustion identifies the original and constructed heights with $g_{M-1}(x)$ and $g_{N-1}(\hat{x})$, respectively, proving the claim. $\square$

For $m \geq 1$, let

$$(5.49) \quad c_m = s_{\min}(aI_m + V_m).$$

By reflection, the two gaps adjacent to the zero eigenvalue of A_{2m+1} have the same height; denote it by

$$(5.50) \quad d_m = \max_{0 \leq x \leq 2r \sin \frac{\pi}{2m+2}} s_{\min}(xI_{2m+1} - A_{2m+1}).$$

LEMMA 5.4 (Central gap comparison). *For every $m \geq 1$,*

$$(5.51) \quad c_m > d_m > c_{m+1}.$$

Proof. Put

$$\zeta = \frac{a + a^{-1}}{2}.$$

For every $j \geq 1$, there is a unique $\rho_j > \cos \frac{\pi}{j+1}$ such that

$$(5.52) \quad aU_j(\rho_j) = U_{j-1}(\rho_j).$$

Indeed, the Christoffel–Darboux identity gives, above the largest zero of U_j ,

$$\frac{d}{dt} \frac{U_{j-1}(t)}{U_j(t)} = -\frac{2 \sum_{k=0}^{j-1} U_k(t)^2}{U_j(t)^2} < 0.$$

The ratio decreases from $+\infty$ to 0, proving existence and uniqueness. A three-term determinant recurrence gives

$$(5.53) \quad \det((aI_j + V_j)^*(aI_j + V_j) - (1 + a^2 - 2at)I_j) = a^j (U_j(t) - a^{-1}U_{j-1}(t)).$$

Thus the largest root ρ_j gives the least eigenvalue. Together with

$$U_j(t)^2 + U_{j-1}(t)^2 - 2tU_j(t)U_{j-1}(t) = 1,$$

equations (5.52) and (5.53) give

$$(5.54) \quad c_j^2 = 1 + a^2 - 2a\rho_j = \frac{1}{U_j(\rho_j)^2}, \quad \zeta - \rho_j = \frac{1}{2U_{j-1}(\rho_j)U_j(\rho_j)}.$$

Next we identify c_m as the exact height of the central gap of A_{2m} . Let

$$P_{\text{oe}} = [\mathbf{e}_1, \mathbf{e}_3, \dots, \mathbf{e}_{2m-1}, \mathbf{e}_2, \mathbf{e}_4, \dots, \mathbf{e}_{2m}]$$

be the permutation matrix that places odd coordinates first. Then

$$P_{\text{oe}}^T A_{2m} P_{\text{oe}} = \begin{bmatrix} 0 & I_m + aV_m^T \\ aI_m + V_m & 0 \end{bmatrix}.$$

Moreover,

$$(I_m + aV_m^T)^*(I_m + aV_m^T) - J_m(aI_m + V_m)^*(aI_m + V_m)J_m = (1 - a^2)\mathbf{e}_m\mathbf{e}_m^T \geq 0.$$

The singular values of the block off-diagonal matrix are the union of the singular values of its two off-diagonal blocks. The Gram inequality shows that $s_{\min}(I_m + aV_m^T) \geq s_{\min}(aI_m + V_m) = c_m$, and hence $s_{\min}(A_{2m}) = c_m$. To identify the maximum throughout the central gap, put $K = 2m + 1$ and use the half variables

$$y = \cos \theta, \quad z = \cosh \eta \quad \text{or} \quad z = \cos \phi.$$

On the positive half of the central middle branch, the preceding chord classification gives

$$(5.55) \quad \mathcal{W}_K(y) = \mathcal{W}_K(z), \quad x = \frac{2r}{\mathfrak{c}_h}yz, \quad s^2 = \frac{4a}{\mathfrak{c}_h^2}(\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2),$$

where $0 \leq y < \cos \frac{m\pi}{K}$ is in the central lobe and z is the endpoint-side outer solution. The function $\mathcal{W}_K$ is even, and strict log-concavity on the central lobe gives

$$\mathcal{W}_K(y) < \mathcal{W}_K(0) = \mathfrak{c}_h \quad (y > 0).$$

Let

$$z_0 = \sqrt{\frac{1 + \rho_m}{2}}.$$

The half-angle identity and (5.52) give

$$U_{2m}(z_0) = U_m(\rho_m) + U_{m-1}(\rho_m) = (1 + a)U_m(\rho_m).$$

Together with (5.54), this yields

$$(5.56) \quad \mathcal{W}_K(z_0) = \mathfrak{c}_h, \quad 4a(\mathfrak{c}_h^2 - z_0^2) = 2a(\zeta - \rho_m) = c_m^2.$$

At $x = 0$, Theorem 5.2 and the chord-exhaustion argument identify the nearest root with c_m , so z_0 is the endpoint-side outer solution of level $\mathfrak{c}_h$. If $x \neq 0$, the level in (5.55) is strictly below $\mathfrak{c}_h$, and hence its endpoint-side solution satisfies $z > z_0$. Equations (5.55)–(5.56) then give

$$s^2 < 4a(\mathfrak{c}_h^2 - z_0^2) = c_m^2.$$

Reflection gives the same strict bound on the other half of the gap. Therefore the central-gap height equals c_m and has the unique maximizer $x = 0$.

Set

$$\omega_m^e = \cos \frac{\pi}{m+1}, \quad \mathcal{V}_m(t) = \sqrt{(\zeta - t)(1 + t)} U_m(t).$$

For the central specialization, take $K = 2m + 2$ and set

$$t_{\text{in}} = 2y^2 - 1 = \cos(2\theta), \quad t_{\text{out}} = 2z^2 - 1.$$

On the positive central gap, $-1 < t_{\text{in}} < -\omega_m^e$, while the endpoint-side outer variable satisfies $\omega_m^e < t_{\text{out}} < \zeta$, where $\zeta = 2c_h^2 - 1$. The identity $U_{2m+1}(t) = 2tU_m(2t^2 - 1)$ gives

$$\begin{aligned} \mathcal{W}_{2m+2}(y)^2 &= (\zeta - t_{\text{in}})(1 + t_{\text{in}})U_m(t_{\text{in}})^2, \\ \mathcal{W}_{2m+2}(z)^2 &= (\zeta - t_{\text{out}})(1 + t_{\text{out}})U_m(t_{\text{out}})^2. \end{aligned}$$

Because $\zeta + 1 = 2c_h^2$, the formulas $x = (2r/c_h)yz$ and $s^2 = (4a/c_h^2)(c_h^2 - y^2)(c_h^2 - z^2)$ become the following chord parametrization:

$$(5.57) \quad \begin{aligned} -1 < t_{\text{in}} < -\omega_m^e, \quad \omega_m^e < t_{\text{out}} < \zeta, \quad \mathcal{V}_m(t_{\text{out}}) &= |\mathcal{V}_m(t_{\text{in}})|, \\ x^2 &= \frac{2a(1 + t_{\text{out}})(1 + t_{\text{in}})}{\zeta + 1}, \quad s^2 = \frac{2a(\zeta - t_{\text{out}})(\zeta - t_{\text{in}})}{\zeta + 1}, \end{aligned}$$

where t_{out} is the descending, or largest, solution. To see this from (5.40), use

$$\mathfrak{s}_h^2 + \sin^2 \theta = \frac{\zeta - t_{\text{in}}}{2}, \quad \mathfrak{s}_h^2 - \sinh^2 \eta = \frac{\zeta - t_{\text{out}}}{2}.$$

Continuation through $t_{\text{out}} = 1$ uses the elliptic coordinate $z = \cos \phi$.

Let $\rho = \rho_m$. The function $\log \mathcal{V}_m$ is strictly concave on (ω_m^e, ζ) . If $(\log \mathcal{V}_m)'(\rho) \geq 0$, then ρ lies at or before the unique lobe maximum. The descending root lies at or after that maximum, so $t_{\text{out}} \geq \rho$. Now suppose $(\log \mathcal{V}_m)'(\rho) < 0$. We claim that $\rho > a$. If $\rho \leq a < 1$, then $\rho < 1$, hence $\rho^2 - 1 \neq 0$, and the following quotient is well defined:

$$\frac{U'_m(\rho)}{U_m(\rho)} = \frac{m\rho - (m+1)a}{\rho^2 - 1}$$

would give

$$\begin{aligned} (\log \mathcal{V}_m)'(\rho) &= \frac{(2m+2)a + 1 - (2m+1)\rho}{2(1 - \rho^2)} - \frac{1}{2(\zeta - \rho)} \\ &\geq \frac{1}{2(1 - \rho)} - \frac{1}{2(\zeta - \rho)} \geq 0, \end{aligned}$$

which contradicts the assumed negative derivative. Hence $\rho > a$. Writing $t_{\text{in}} = -\cos \phi_c$, $0 < \phi_c < \pi/(m+1)$, gives

$$\begin{aligned} |\mathcal{V}_m(t_{\text{in}})|^2 &= \frac{(\zeta + \cos \phi_c) \sin^2((m+1)\phi_c)}{1 + \cos \phi_c} \\ &< \frac{\zeta + \omega_m^e}{1 + \omega_m^e} < \frac{1 + a}{2a} < \frac{1 + \rho}{2a} = \mathcal{V}_m(\rho)^2. \end{aligned}$$

Now ρ is on the descending side and $|\mathcal{V}_m(t_{\text{in}})| < \mathcal{V}_m(\rho)$, so the unique descending root again satisfies $t_{\text{out}} > \rho$. In either case, $t_{\text{out}} \geq \rho$, and $t_{\text{in}} > -1$ makes the following inequality strict:

$$s^2 < 2a(\zeta - \rho) = c_m^2.$$

This holds along the whole central sheet, so $d_m < c_m$. Thus the odd-order central height lies strictly below the preceding even-order central height. To complete the alternating comparison, it remains to place it strictly above the next even-order height.

Lower comparison $d_m > c_{m+1}$: decisive scalar estimate

Put $L = m + 1 \geq 2$, and abbreviate

$$\rho = \rho_L, \quad u_- = U_{L-1}(\rho), \quad u_+ = U_L(\rho), \quad \omega_L = \cos \frac{\pi}{2L}, \quad \Delta_\omega = 1 - \omega_L,$$

$$(5.58) \quad \mathfrak{D}_L = 1 + 2(\rho + \omega_L)u_-u_+, \quad \xi = \rho - \frac{\Delta_\omega}{\mathfrak{D}_L}.$$

Thus $a = u_-/u_+$ and $c_L = 1/u_+$.

The required inequality is

$$(5.59) \quad (1 + \omega_L)(1 + \xi)^2 U_{L-1}(\xi)^2 < (\rho + \omega_L)\mathfrak{D}_L.$$

[Appendix E](#) proves (5.59) for every $L \geq 2$ from the Chebyshev identity and strict log-concavity, treating the three cases $\rho > 1$, $\rho = 1$, and $\rho < 1$.

Certification of the least singular value

Since $a > 0$, $\Delta_\omega > 0$, $\rho + \omega_L > 0$, and $\zeta + \omega_L > 0$, let $x_* > 0$ be the positive square root determined by

$$(5.60) \quad x_*^2 = \frac{2a\Delta_\omega(\rho + \omega_L)}{\zeta + \omega_L}, \quad \tau_* = c_L.$$

The algebraic certification is given in [appendix F](#). Substitution of the distinct folded pair $u = \xi$, $v = -\omega_L$ into (5.36), followed by (5.59), proves

$$(5.61) \quad q_{2L-1}(x_*, \tau_*)q_{2L-1}(x_*, -\tau_*) > 0.$$

At $x = 0$, the second-smallest singular value of A_{2L-1} is

$$s_{L,2}^{(0)} = \sqrt{1 + a^2 - 2a \cos \frac{\pi}{L}}.$$

The same appendix proves the strict separation

$$(5.62) \quad x_* + \tau_* < 2\sqrt{a} \sin \frac{\pi}{2L} \leq s_{L,2}^{(0)}.$$

The product parity and this second-singular-value separation now identify the relevant signed root independently of any prior branch choice.

Let $\mu_1, \dots, \mu_{2L-1}$ be the eigenvalues of the real symmetric matrix

$$B_{2L-1}(x_*) = (x_* I_{2L-1} - A_{2L-1})J_{2L-1}.$$

Right multiplication by J_{2L-1} preserves singular values, so the numbers $|\mu_i|$ are precisely the singular values of $x_* I_{2L-1} - A_{2L-1}$. Moreover,

$$q_{2L-1}(x_*, \tau_*)q_{2L-1}(x_*, -\tau_*) = \prod_{i=1}^{2L-1} (\mu_i^2 - \tau_*^2).$$

Positivity of the product in (5.61) implies that the number of indices with $|\mu_i| < \tau_*$ is even. On the other hand, Weyl's singular-value perturbation inequality and (5.62) give, for the second-smallest singular value s_{next} ,

$$s_{\text{next}}(x_* I_{2L-1} - A_{2L-1}) \geq s_{L,2}^{(0)} - x_* > \tau_*,$$

so at most one index can satisfy $|\mu_i| < \tau_*$. Parity therefore gives $|\mu_i| \geq \tau_*$ for every i , and strict positivity of the product rules out equality. Hence $s_{\min}(x_* I_{2L-1} - A_{2L-1}) > c_L$. Since $0 < x_* < 2r \sin \frac{\pi}{2L}$, the point lies strictly in the positive central gap, and $d_{L-1} > c_L$. Together with the upper comparison $c_m > d_m$, this proves (5.51). $\square$

Proof of Theorem 5.1. Reflection and the central interlacing chain now assemble the local comparisons into the global barrier inequality. Put $N = n + 1$ and $M = N + 1$, so the larger and smaller matrices have orders $M - 1$ and $N - 1$, respectively. If $\Sigma = \text{diag}(1, -1, 1, -1, \dots)$, then $\Sigma A_n \Sigma = -A_n$, so reflection reduces every noncentral comparison to a positive gap. For each positive noncentral gap $G_{M-1,j}$, choose $x_j \in G_{M-1,j}^\circ$ attaining $\gamma_{M-1,j}$. Then $j \geq 1$ and $(j+1)\pi/N \leq \pi/2$, so Theorem 5.3 produces a point $\hat{x}_j \in G_{N-1,j}^\circ$, and chord exhaustion gives

$$\gamma_{N-1,j} \geq g_{N-1}(\hat{x}_j) > g_{M-1}(x_j) = \gamma_{M-1,j}.$$

Reflection gives the same inequality on the corresponding reflected gap.

By Theorem 5.4, if $M = 2\ell_* + 1$, the remaining single central height of the larger, even-order matrix is c_{ℓ_*} , whereas each central height of the smaller odd-order matrix is $d_{\ell_*-1} > c_{\ell_*}$. If $M = 2\ell_* + 2$, the two central heights of the larger odd-order matrix equal $d_{\ell_*} < c_{\ell_*}$, where c_{ℓ_*} is the single central height of the smaller even-order matrix. Thus every height of the larger problem is strictly smaller than at least one height of the smaller problem. Choose j_* with $\gamma_{M-1,j_*} = \max_j \gamma_{M-1,j}$. Then

$$\gamma_{M-1} = \gamma_{M-1,j_*} < \max_k \gamma_{N-1,k} = \gamma_{N-1}.$$

Since $M - 1 = n + 1$ and $N - 1 = n$, this proves $\gamma_{n+1} < \gamma_n$. $\square$

6. Quantitative bounds and the transition scale. The connectedness criterion (4.25) and the strict comparison of Theorem 5.1 reduce the remaining problem to estimating γ_n . Put $N = n + 1$ and recall $\mathcal{L}_n(a)$ and $\mathcal{U}_n(a)$ from (3.16)–(3.17).

PROPOSITION 6.1 (Sharp-order gap bounds). *For every $n \geq 2$,*

$$(6.63) \quad \mathcal{L}_n(a) \leq \gamma_n \leq \mathcal{U}_n(a).$$

In particular, both bounds have order $(n+1)a^{n/2}$, with constants depending only on a .

Proof. For $x = 2r \cos \theta \in I_n^{\text{sp}}$, set

$$\mathbf{v} = (v_j)_{j=1}^n, \quad v_j = r^{j-1} \sin(j\theta), \quad 1 \leq j \leq n.$$

The residual $(xI_n - A_n)\mathbf{v}$ is supported in its last coordinate, where it equals $r^n \sin(N\theta)$. Thus

$$g_n(2r \cos \theta) \leq \frac{r^n |\sin(N\theta)|}{\left(\sum_{j=1}^n r^{2j-2} \sin^2(j\theta)\right)^{1/2}} \leq r^n \frac{|\sin(N\theta)|}{\sin \theta}.$$

Since $\pi/N \leq \theta \leq (N-1)\pi/N$, this proves the upper bound.

For the lower bound, put $p_\theta = re^{i\theta}$, $q_\theta = re^{-i\theta}$, so that $p_\theta + q_\theta = x$ and $p_\theta q_\theta = a$. For $\mathbf{u} = (u_j)_{j=1}^n \in \mathbb{C}^n$, put $u_0 = u_{n+1} = 0$, and define maps from $\mathbb{C}^n$ to $\mathbb{C}^{n+1}$ by

$$(\mathbf{C}_\theta \mathbf{u})_j = p_\theta u_j - u_{j+1}, \quad (\mathbf{D}_\theta \mathbf{u})_j = u_j - \overline{q_\theta} u_{j+1}, \quad 0 \leq j \leq n.$$

Direct multiplication, using $p_\theta + q_\theta = x$ and $p_\theta q_\theta = a$, gives

$$(6.64) \quad xI_n - A_n = \mathbf{D}_\theta^* \mathbf{C}_\theta.$$

The two Gram matrices are diagonally unitarily similar to the tridiagonal Toeplitz matrix with diagonal $1 + r^2$ and off-diagonal $-r$. Its eigenvalues are $1 + r^2 - 2r \cos \frac{k\pi}{N}$, $1 \leq k \leq n$. Consequently, for every $\mathbf{u} \in \mathbb{C}^n$,

$$(6.65) \quad \begin{aligned} \|\mathbf{C}_\theta \mathbf{u}\|_2 &\geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right)^{1/2} \|\mathbf{u}\|_2, \\ \|\mathbf{D}_\theta \mathbf{u}\|_2 &\geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right)^{1/2} \|\mathbf{u}\|_2. \end{aligned}$$

Put

$$\mathcal{S}_N = \sum_{j=0}^n r^{2j} = \frac{1 - r^{2N}}{1 - r^2}.$$

Unit normal vectors to the two hyperplane ranges are

$$\mathbf{u}_C = \mathcal{S}_N^{-1/2} (\overline{p_\theta}^n, \overline{p_\theta}^{n-1}, \dots, 1)^T, \quad \mathbf{u}_D = \mathcal{S}_N^{-1/2} (1, q_\theta, \dots, q_\theta^n)^T.$$

In the polar decompositions

$$\mathbf{C}_\theta = Q_C (\mathbf{C}_\theta^* \mathbf{C}_\theta)^{1/2}, \quad \mathbf{D}_\theta = Q_D (\mathbf{D}_\theta^* \mathbf{D}_\theta)^{1/2},$$

the matrices Q_C, Q_D are isometries onto these ranges. Two hyperplanes in $\mathbb{C}^{n+1}$ intersect in dimension $n - 1$, so the cross-Gram matrix $Q_D^* Q_C$ has $n - 1$ singular values equal to 1; its final singular value is $|\langle \mathbf{u}_D, \mathbf{u}_C \rangle|$, the cosine of the nontrivial principal angle. The geometric sum gives

$$(6.66) \quad \begin{aligned} |\langle \mathbf{u}_D, \mathbf{u}_C \rangle| &= \frac{r^n}{\mathcal{S}_N} \left| \sum_{j=0}^{N-1} e^{i(2j - (N-1))\theta} \right| \\ &= \frac{r^n(1 - r^2)}{1 - r^{2N}} \left| \frac{\sin(N\theta)}{\sin \theta} \right| =: \chi_N(\theta). \end{aligned}$$

Finally,

$$xI_n - A_n = (\mathbf{D}_\theta^* \mathbf{D}_\theta)^{1/2} Q_D^* Q_C (\mathbf{C}_\theta^* \mathbf{C}_\theta)^{1/2}.$$

Using $s_{\min}(XYZ) \geq s_{\min}(X)s_{\min}(Y)s_{\min}(Z)$ and (6.65) gives

$$g_n(2r \cos \theta) \geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right) \chi_N(\theta).$$

Putting $\theta = 3\pi/(2N)$, which lies in the first spectral gap, proves the lower bound.

Finally,

$$\mathcal{U}_n(a) \leq \frac{1}{2}(n+1)r^n,$$

while

$$(6.67) \quad \mathcal{L}_n(a) \geq C_{\text{lb}}(a)(n+1)r^n, \quad C_{\text{lb}}(a) = \frac{2(1-r)^2(1-r^2)}{3\pi}.$$

These follow from $\sin \frac{\pi}{N} \geq 2/N$, $1+r^2-2r \cos \frac{\pi}{N} \geq (1-r)^2$, and $\sin \frac{3\pi}{2N} \leq 3\pi/(2N)$. This proves the asserted sharp order. $\square$

Proof of Theorem 3.3. By Theorem 6.1, $\gamma_n \leq \mathcal{U}_n(a) \rightarrow 0$. Theorem 5.1 says that γ_n is strictly decreasing. Equation (4.25) therefore shows that, once the pseudospectrum is connected, it stays connected in every larger dimension, which proves the set identity in part 3. Contractibility and the connectedness criterion are exactly (4.25).

If $\mathcal{L}_n(a) \geq \varepsilon$, then $\gamma_n \geq \varepsilon$, so the pseudospectrum in dimension n is disconnected. If $\mathcal{U}_n(a) < \varepsilon$, then $\gamma_n < \varepsilon$, so it is connected. Strict decrease of γ_n now gives (3.18).

For $t > 0$, define

$$\mathfrak{e}_n = (n+1)r^n, \quad \mathcal{T}_r(t) = \min\{N_0 \geq 2 : \mathfrak{e}_m < t \text{ for every } m \geq N_0\}.$$

This minimum exists because $\mathfrak{e}_m \rightarrow 0$. Because persistence gives

$$N_c(a, \varepsilon) = \min\{N \geq 2 : \gamma_m < \varepsilon \text{ for every } m \geq N\},$$

the bounds $C_{\text{lb}}(a)\mathfrak{e}_m \leq \gamma_m \leq \mathfrak{e}_m/2$ immediately give

$$(6.68) \quad \mathcal{T}_r(\varepsilon/C_{\text{lb}}(a)) \leq N_c(a, \varepsilon) \leq \mathcal{T}_r(2\varepsilon).$$

Thus the polynomial factor $n+1$, as well as the exponential factor $a^{n/2}$, is sharp on both sides.

Elementary inversion of the envelope threshold

Let

$$\beta = -\log r = \frac{1}{2} \log(1/a), \quad L(t) = \log(1/t), \quad S_r(t) = \frac{L(t) + \log L(t)}{\beta}.$$

For every fixed $D \in \mathbb{R}$, direct substitution gives

$$\beta(S_r(t) + D) - \log(S_r(t) + D + 1) - L(t) = \beta D + \log \beta + o(1) \quad (t \downarrow 0).$$

Indeed, $\beta S_r = L + \log L$ and $\log(S_r + D + 1) = \log L - \log \beta + o(1)$. Choose constants $D_- < -(\log \beta)/\beta < D_+$. For all sufficiently small t , the preceding residual is negative at $S_r(t) + D_-$ and positive at $S_r(t) + D_+$. Since $S_r(t) \rightarrow \infty$ and

$$\frac{d}{dx}((x+1)e^{-\beta x}) = e^{-\beta x}(1 - \beta(x+1)),$$

this function is decreasing on $[S_r(t) + D_-, \infty)$ for all sufficiently small t . The strict integer tail condition therefore gives

$$\lfloor S_r(t) + D_- \rfloor < \mathcal{T}_r(t) \leq \lceil S_r(t) + D_+ \rceil.$$

Thus $\mathcal{T}_r(t) = S_r(t) + O_r(1)$. Replacing t by a fixed positive multiple changes $S_r(t)$ by only $O_r(1)$; applying this to the two sides of (6.68) proves (3.19).

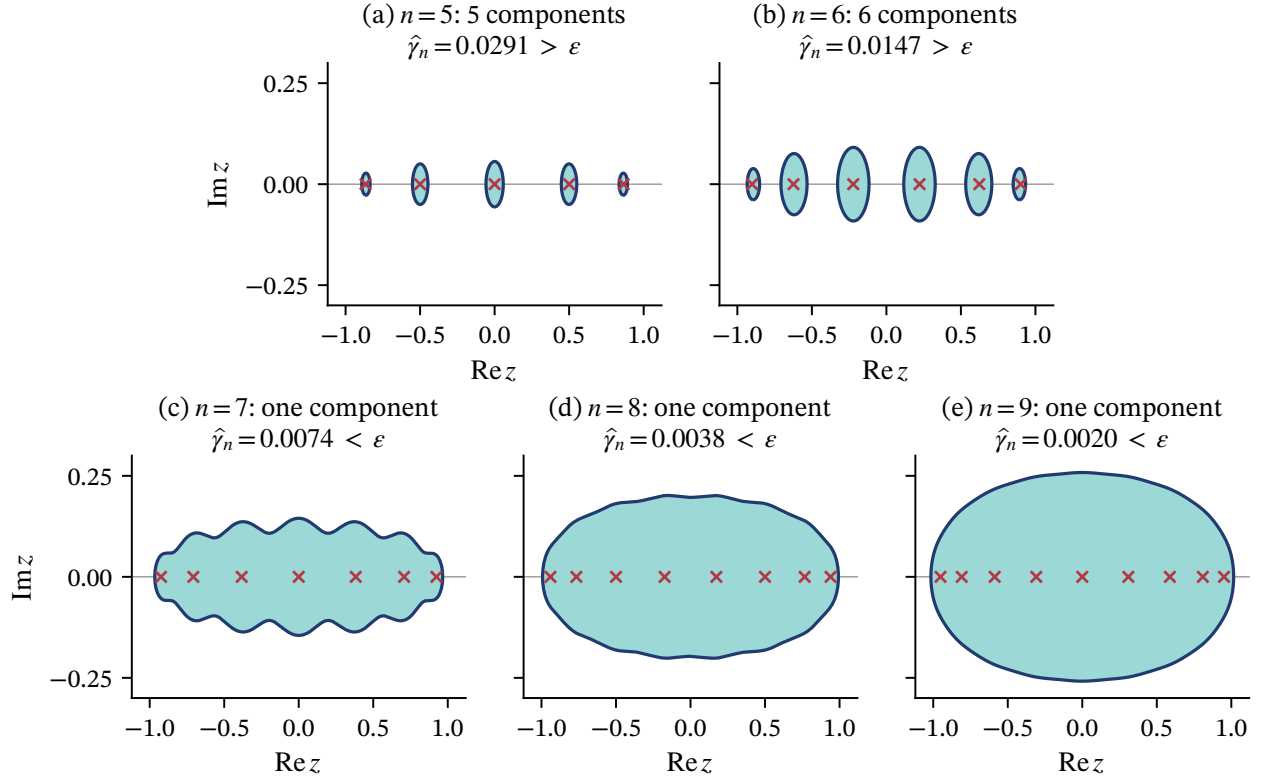


FIGURE 1. Numerically sampled strict pseudospectra of $A_n(1/4)$ at $\varepsilon = 10^{-2}$. Shading marks the sampled pseudospectrum and crosses mark the exact eigenvalues. The samples show component coalescence between $n = 6$ and $n = 7$.

Exact two-site endpoint

For $|x| \leq r$,

$$\det((xI_2 - A_2)^*(xI_2 - A_2) - a^2 I_2) = x^2(x^2 - 2a - 2a^2) \leq 0.$$

The matrix inside the determinant is Hermitian, and the nonpositive product of its two eigenvalues forces its least eigenvalue to be nonpositive. Thus $s_{\min}(xI_2 - A_2) \leq a$. At $x = 0$, the singular values of $-A_2$ are 1 and a , so equality is attained and $\gamma_2 = a$. Hence (4.25) gives $N_c(a, \varepsilon) = 2$ if and only if $\varepsilon > a$. This proves all assertions. $\square$

7. Numerical illustrations. Figures 1 and 2 illustrate two aspects of the matrix-order transition: coalescence of pseudospectral components and decay of the real-axis backward-error barrier.

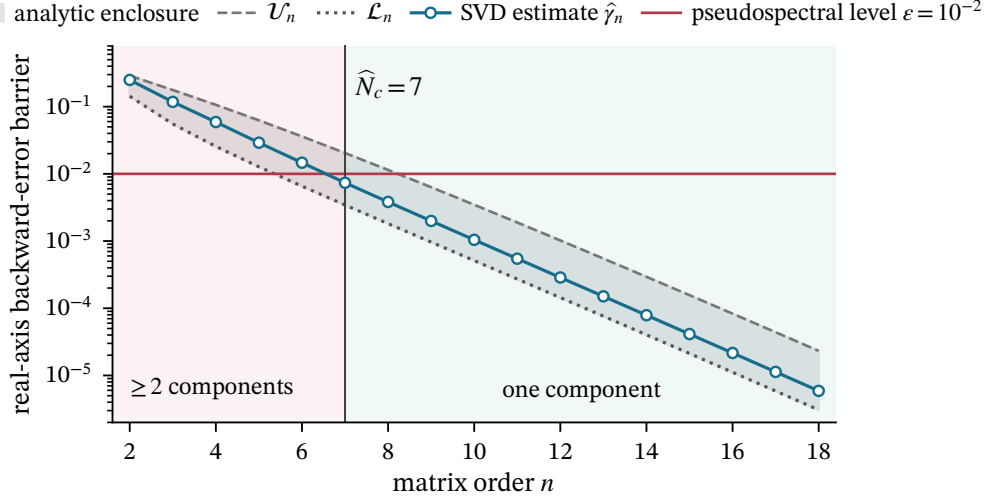


FIGURE 2. Real-axis backward-error barriers for $a = 1/4$. The grey band, bounded by the dotted lower curve $\mathcal{L}_n(a)$ and dashed upper curve $\mathcal{U}_n(a)$, is the rigorous enclosure of γ_n and proves the scale $(n+1)a^{n/2}$. Blue circles joined by a solid line are floating-point estimates; the red horizontal line is the fixed pseudospectral level $\varepsilon = 10^{-2}$, and the black vertical line marks the numerical crossing $\hat{N}_c = 7$. The theorem gives the analytic bracket $6 \leq N_c \leq 9$.

8. Discussion. The classification separates three genuinely different finite-size mechanisms. In the irreducible nonnormal regime, the exact threshold has the exponential–polynomial scale $c(n+1)a^{n/2}$, and the first connected order grows logarithmically in $1/\varepsilon$. At the normal boundary, the threshold is the largest half-gap, has scale $\pi c/(n+1)$, and the critical order grows as $1/\varepsilon$. At the reducible boundary, the pseudospectrum is already a disk and the threshold is zero. The two boundaries are singular limits of the interior scale: the remainder $O_a(1)$ in (3.19) is pointwise for fixed $0 < a < 1$.

This trichotomy also explains the role of strict dimension monotonicity. For every irreducible complex tridiagonal Toeplitz family, normal or nonnormal, the final barrier decreases at each order, so the connected orders form one integer tail. At reducibility the barrier collapses to zero at every order, and persistence holds without a strict inequality. The statement is topological rather than merely spectral: pseudospectral components are robust spectral clusters under an unstructured norm bound, and N_T is the first order at which all such clusters have coalesced.

The pseudospectrum throughout is generated by the full complex spectral-norm ball. Restricting perturbations to Toeplitz, tridiagonal, real, or entrywise patterns changes the attainable set and may change its connectedness threshold; see [7] for Toeplitz-structured pseudospectra. Extending the strict order comparison to such structured sets, to block Toeplitz matrices, or to periodic k -Toeplitz families provides natural directions for further study.

Restricting the nearest-neighbor Hatano–Nelson hopping law [8, 9] to an open finite chain and, if necessary, reversing its orientation gives, up to sign and scale, $A_n(a)$, with $t_L \geq t_R > 0$ and $a = t_R/t_L$. Its reciprocal case is the normal boundary, its one-way limit is the Jordan boundary, and its nonreciprocal interior is the nonnormal regime. Writing $h = \frac{1}{2} \log(t_L/t_R)$, (3.15) gives the familiar right-eigenvector envelope $e^{-h(j-1)}$ [2]; the same exponent appears in $a^{n/2} = e^{-hn}$. This interpretation motivates the three mathematical regimes within a standard nonreciprocal-chain model.

In summary, the general vertical theorem identifies the topological mechanism, the unitary–affine reduction classifies the full complex family, and the canonical comparison supplies the nontrivial all-order law. The result is an exact finite-dimensional classification of connectedness, including the boundary levels.

Appendix A. Pentadiagonal continuant certificate for vertical monotonicity.

PROPOSITION A.1 (Vertical determinant certificate). *Fix $0 < a < 1$, $x \in \mathbb{R}$, $Y \geq 0$, and $t \in \mathbb{R}$, and extend the matrix notation by $A_1 = [0]$. For $k \geq 1$, set*

$$\begin{aligned}\mathcal{P}_k(Y, t) &= ((x + i\sqrt{Y})I_k - A_k)^*((x + i\sqrt{Y})I_k - A_k) - t^2 I_k, \\ F_k(Y, t) &= \det \mathcal{P}_k(Y, t),\end{aligned}$$

and set $F_0 = 1$. For every $n \geq 1$,

$$\mathcal{P}_n(Y, t) > 0 \implies \partial_Y F_n(Y, t) > 0.$$

Proof. At a fixed value of Y , write

$$\delta_{\text{pent}} = x^2 + Y + 1 + a^2 - t^2, \quad b = -(1 + a)x + i(1 - a)\sqrt{Y}.$$

For $n \geq 2$, the matrix $\mathcal{P}_n(Y, t)$ has diagonal

$$\delta_{\text{pent}} - 1, \delta_{\text{pent}}, \dots, \delta_{\text{pent}}, \delta_{\text{pent}} - a^2,$$

whereas for $n = 1$ its sole entry is $\delta_{\text{pent}} - 1 - a^2$. In all dimensions its first upper diagonal is b , its first lower diagonal is $\bar{b}$, and its second off-diagonals are a . Let T_k be the determinant of the pure pentadiagonal Toeplitz matrix with diagonal δ_{pent} and these two off-diagonals. Set $T_{-1} = 0, T_0 = 1$, and put

$$(A.69) \quad D_k = T_k - T_{k-1}, \quad R_k = T_k - a^2 T_{k-1}.$$

Indeed, a determinant is affine in each diagonal entry. Replacing the first diagonal entry δ_{pent} by $\delta_{\text{pent}} - 1$ subtracts the cofactor T_{k-1} , whereas replacing the last one by $\delta_{\text{pent}} - a^2$ subtracts $a^2 T_{k-1}$. Thus, for $k < n$, D_k and R_k are, respectively, the leading and trailing principal minors of $\mathcal{P}_n(Y, t)$.

The following continuant calculation isolates all dependence on Y . With $T_k = 0$ for $k < 0$, a Laplace expansion over the last two rows, grouped according to the largest displaced index in the contributing permutation, gives for every $k \geq 1$

$$(A.70) \quad \begin{aligned}T_k &= \delta_{\text{pent}} T_{k-1} - (|b|^2 - a^2) T_{k-2} - (2a^2 \delta_{\text{pent}} - a(b^2 + \bar{b}^2)) T_{k-3} \\ &\quad - a^2(|b|^2 - a^2) T_{k-4} + a^4 \delta_{\text{pent}} T_{k-5} - a^6 T_{k-6} - a^2 \mathbf{1}_{\{k=2\}}.\end{aligned}$$

For example, $T_1 = \delta_{\text{pent}}$ and $T_2 = \delta_{\text{pent}}^2 - |b|^2$; these also verify the only inhomogeneous coefficient in (A.70). Multiplying the recurrence by z^k and summing gives

$$(A.71) \quad \sum_{k \geq 0} T_k z^k = \frac{1 - a^2 z^2}{Q(z, Y)},$$

where

$$\begin{aligned}Q(z, Y) &= 1 - \delta_{\text{pent}} z + (|b|^2 - a^2) z^2 + (2a^2 \delta_{\text{pent}} - a(b^2 + \bar{b}^2)) z^3 \\ &\quad + a^2(|b|^2 - a^2) z^4 - a^4 \delta_{\text{pent}} z^5 + a^6 z^6.\end{aligned}$$

Consequently, with $\mathcal{D}(z, Y) = \sum_{k \geq 0} D_k z^k$,

$$(A.72) \quad \mathcal{D}(z, Y) = \frac{(1-z)(1-a^2 z^2)}{Q(z, Y)}, \quad \sum_{k \geq 0} F_k(Y, t) z^k = (1-a^2 z) \mathcal{D}(z, Y).$$

For the second identity, first change the initial diagonal, producing D_k , and then change the final diagonal; its cofactor is D_{k-1} . Thus $F_k = D_k - a^2 D_{k-1}$, which proves the asserted generating function. Direct substitution of δ_{pent} and b gives

$$\partial_Y Q = -z(1-z)(1+az)^2(1-a^2 z).$$

Differentiating (A.72) therefore yields

$$(A.73) \quad \sum_{k \geq 0} \partial_Y F_k(Y, t) z^k = \varphi_a(z) \mathcal{D}(z, Y)^2, \quad \varphi_a(z) = \frac{z(1+az)(1-a^2 z)^2}{1-az}.$$

Suppose that $\mathcal{P}_n(Y, t) > 0$. Then

$$(A.74) \quad D_k > 0, \quad R_k > 0 \quad (0 \leq k < n).$$

Put $u_k = \sum_{j=0}^k D_j D_{k-j}$, with $u_k = 0$ for $k < 0$. The coefficients of φ_a , beginning with the coefficient of z , are

$$1, \quad 2a(1-a), \quad a^2(a^2-4a+2), \quad 2a^{k-1}(1-a)^2 \quad (k \geq 4).$$

Thus

$$(A.75) \quad \begin{aligned} \partial_Y F_n &= u_{n-1} + 2a(1-a)u_{n-2} + a^2(a^2-4a+2)u_{n-3} \\ &\quad + 2(1-a)^2 \sum_{k=4}^n a^{k-1} u_{n-k}. \end{aligned}$$

The third coefficient is the sole term that may be negative; the following cofactor inequality controls its contribution. Let $j, k \geq 1$, $j+k = n-1$. Deleting index $k+1$ from $\mathcal{P}_n$ leaves blocks of sizes k and j , with the second off-diagonal a as their sole coupling. In block form, the deleted principal submatrix is

$$\begin{bmatrix} L_k & a\mathbf{e}_k \mathbf{e}_1^T \\ a\mathbf{e}_1 \mathbf{e}_k^T & R_j^{\text{mat}} \end{bmatrix}, \quad \det L_k = D_k, \quad \det R_j^{\text{mat}} = R_j.$$

Expanding once in each of the two coupling entries shows that its determinant is

$$(A.76) \quad D_k R_j - a^2 D_{k-1} R_{j-1} > 0.$$

Deleting index $j+1$ instead gives the same inequality with j, k interchanged. Define

$$\vartheta_k = \frac{D_k}{D_{k-1}}, \quad \varpi_k = \frac{R_k}{R_{k-1}}.$$

Since $T_k = \sum_{\ell=0}^k D_\ell$, (A.69) gives

$$(A.77) \quad R_k = D_k + (1-a^2) \sum_{\ell=0}^{k-1} D_\ell, \quad R_k - R_{k-1} = D_k - a^2 D_{k-1}.$$

Hence $\vartheta_k \leq a^2$ implies $\varpi_k \leq 1$. On dividing (A.76) and its companion by positive factors, we obtain

$$(A.78) \quad \vartheta_k \varpi_j > a^2, \quad \vartheta_j \varpi_k > a^2.$$

If both ϑ_j and ϑ_k were at most a^2 , then both ϖ_j and ϖ_k would be at most 1, contradicting (A.78). Hence, after interchanging j, k if necessary, $\vartheta_j > a^2$. Moreover,

$$\vartheta_j R_{j-1} - a^2 R_j = (\vartheta_j - a^2)(R_{j-1} - a^2 D_{j-1}),$$

and the second factor equals $(1 - a^2) \sum_{\ell=0}^{j-1} D_\ell > 0$. Therefore

$$\vartheta_j > a^2 \implies \varpi_j < \vartheta_j / a^2.$$

The first inequality in (A.78) now gives $a^2 < \vartheta_k \varpi_j < \vartheta_j \vartheta_k / a^2$, and therefore

$$(A.79) \quad \vartheta_j \vartheta_k > a^4.$$

Consequently, for $n \geq 3$, applying (A.79) to $j = p + 1$, $k = n - 2 - p$ gives

$$D_{p+1} D_{n-2-p} > a^4 D_p D_{n-3-p}, \quad 0 \leq p \leq n - 3.$$

Summing and then adding the two positive endpoint terms gives

$$(A.80) \quad u_{n-1} = 2D_{n-1} + \sum_{j=1}^{n-2} D_j D_{n-1-j} > a^4 u_{n-3}.$$

If the third coefficient in (A.75) is negative, write it as $-\mathcal{B}_{\text{neg}}$, where $\mathcal{B}_{\text{neg}} = a^2(4a - 2 - a^2)$. Since

$$a^4 - \mathcal{B}_{\text{neg}} = 2a^2(1 - a)^2 > 0,$$

equations (A.75) and (A.80) prove (4.24). If that coefficient is nonnegative, the implication is immediate. For $n = 1$, one has $\partial_Y F_1 = 1$; for $n = 2$, one has $\partial_Y F_2 = 2D_1 + 2a(1 - a) > 0$. Thus (4.24) holds in every dimension, which proves the proposition. $\square$

Appendix B. Folded-continuant transfer and boundary residuals.

This appendix supplies the transfer and generating identities underlying (5.35)–(5.36) and records every boundary sign and parity convention in the signed-pencil argument. Pair the coordinates in the order $(1, n), (2, n - 1), \dots$. In this order the symmetric matrix $(xI_n - A_n)J_n - sI_n$ is block tridiagonal. Its interior diagonal and off-diagonal blocks are

$$D_{\text{int}} = \begin{bmatrix} -s & x \\ x & -s \end{bmatrix}, \quad E_{\text{off}} = \begin{bmatrix} 0 & -1 \\ -a & 0 \end{bmatrix},$$

with lower block E_{off}^T . For even order the terminal block is

$$D_{\text{term}}^e = \begin{bmatrix} -1 - s & x \\ x & -a - s \end{bmatrix};$$

for odd order the terminal block is the scalar $x - s$, coupled to the preceding block by the column $(-1, -a)^T$. Since

$$(xI_n - A_n - sJ_n) = ((xI_n - A_n)J_n - sI_n)J_n,$$

the determinant of this block matrix, multiplied by $\det J_n$, is $q_n(x, s)$. Set $\mathcal{Q}_m^e = q_{2m}(x, s)$ and $\mathcal{Q}_m^o = q_{2m+1}(x, s)$. The following five-minor transfer completes the continuant verification. For $\tau \in \{e, o\}$, let K_m^τ denote the corresponding matrix

$$(xI_n - A_n - sJ_n)J_n = (xI_n - A_n)J_n - sI_n, \quad n = 2m \text{ or } 2m + 1,$$

after simultaneously permuting its rows and columns into the paired order. Thus K_m^e has order $2m$, K_m^o has order $2m + 1$, and

$$K_1^e = \begin{bmatrix} -1-s & x \\ x & -a-s \end{bmatrix}, \quad K_1^o = \begin{bmatrix} -s & x & -1 \\ x & -s & -a \\ -1 & -a & x-s \end{bmatrix}.$$

For $m \geq 1$, increasing m by one adds one interior block at the exposed end:

$$(B.81) \quad K_{m+1}^\tau = \begin{bmatrix} D_{\text{int}} & \mathcal{E}_m^\tau \\ (\mathcal{E}_m^\tau)^T & K_m^\tau \end{bmatrix}, \quad \mathcal{E}_m^\tau = \begin{bmatrix} E_{\text{off}} & 0 \end{bmatrix},$$

where the zero block has the width required by K_m^τ .

For a matrix whose first two coordinates are exposed, write $K(\widehat{I} \mid \widehat{J})$ for the matrix obtained by deleting rows I and columns J . Suppressing the parity superscript, set

$$\begin{aligned} \mathfrak{d}_m &= \det K_m, & \mathfrak{p}_m &= \det K_m(\widehat{1} \mid \widehat{1}), & \mathfrak{q}_m &= \det K_m(\widehat{2} \mid \widehat{2}), \\ \mathfrak{c}_m &= \det K_m(\widehat{1} \mid \widehat{2}), & \mathfrak{r}_m &= \det K_m(\widehat{1}, \widehat{2} \mid \widehat{1}, \widehat{2}). \end{aligned}$$

The off-principal minor obtained by deleting row 2 and column 1 has the same value $\mathfrak{c}_m$, because K_m is symmetric.

The one-step transfer follows from a Schur complement. On the dense set where both K_m and the tail obtained by deleting its first two coordinates are invertible, let

$$F = \begin{bmatrix} u & w \\ w & v \end{bmatrix}$$

be its Schur complement in K_m , and let δ be the determinant of that tail. Then

$$(\mathfrak{d}_m, \mathfrak{p}_m, \mathfrak{q}_m, \mathfrak{c}_m, \mathfrak{r}_m) = \delta(uv - w^2, v, u, w, 1).$$

Since

$$E_{\text{off}} \text{adj}(F) E_{\text{off}}^T = \begin{bmatrix} u & -aw \\ -aw & a^2v \end{bmatrix},$$

taking the Schur complement of K_m in (B.81) gives

$$(B.82) \quad \begin{aligned} \mathfrak{d}_{m+1} &= (s^2 - x^2)\mathfrak{d}_m + s(a^2\mathfrak{p}_m + \mathfrak{q}_m) - 2ax\mathfrak{c}_m + a^2\mathfrak{r}_m, \\ \mathfrak{p}_{m+1} &= -s\mathfrak{d}_m - a^2\mathfrak{p}_m, \\ \mathfrak{q}_{m+1} &= -s\mathfrak{d}_m - \mathfrak{q}_m, \\ \mathfrak{c}_{m+1} &= x\mathfrak{d}_m + a\mathfrak{c}_m, \\ \mathfrak{r}_{m+1} &= \mathfrak{d}_m. \end{aligned}$$

Both sides are polynomial in x and s . For each fixed $0 < a < 1$, the identities hold on a dense nonsingular set, so the polynomial identity theorem extends them to all x, s , including singular matrix cases.

The five quantities in (B.82) satisfy one linear invariant. Define

$$\ell_m = (1-a)\mathfrak{d}_m - as\mathfrak{p}_m + s\mathfrak{q}_m + (1-a)x\mathfrak{c}_m - a(1-a)\mathfrak{r}_m.$$

Direct substitution into (B.82) gives

$$\ell_{m+1} = -a\ell_m.$$

For the two terminal blocks, direct evaluation gives

$$\begin{aligned} & (\mathfrak{d}_1, \mathfrak{p}_1, \mathfrak{q}_1, \mathfrak{c}_1, \mathfrak{r}_1)_e \\ &= ((1+s)(a+s) - x^2, -a-s, -1-s, x, 1), \\ & (\mathfrak{d}_1, \mathfrak{p}_1, \mathfrak{q}_1, \mathfrak{c}_1, \mathfrak{r}_1)_o \\ &= (x(1+a^2) + 2as - C_0(x-s), \\ & \quad s^2 - sx - a^2, s^2 - sx - 1, x^2 - sx - a, x-s). \end{aligned}$$

Thus $\ell_1 = 0$ in both parity classes, and consequently

$$(B.83) \quad (1-a)\mathfrak{d}_m - as\mathfrak{p}_m + s\mathfrak{q}_m + (1-a)x\mathfrak{c}_m - a(1-a)\mathfrak{r}_m = 0 \quad (m \geq 1).$$

Because

$$\det J_{2m} = \det J_{2m+1} = (-1)^m,$$

the first coordinate of the following signed state is the required determinant:

$$\boldsymbol{\eta}_m^\tau = (-1)^m [\mathfrak{d}_m^\tau \quad \mathfrak{p}_m^\tau \quad \mathfrak{q}_m^\tau \quad \mathfrak{c}_m^\tau]^T, \quad (\boldsymbol{\eta}_m^e)_1 = \mathcal{Q}_m^e, \quad (\boldsymbol{\eta}_m^o)_1 = \mathcal{Q}_m^o.$$

Eliminating $\mathfrak{r}_m$ with (B.83) reduces (B.82) to the four-dimensional transfer

$$(B.84) \quad \boldsymbol{\eta}_{m+1}^\tau = \mathsf{T}_{\text{fold}} \boldsymbol{\eta}_m^\tau, \quad \mathsf{T}_{\text{fold}} = \begin{bmatrix} x^2 - s^2 - a & \frac{a^3 s}{1-a} & -\frac{s}{1-a} & ax \\ s & a^2 & 0 & 0 \\ s & 0 & 1 & 0 \\ -x & 0 & 0 & -a \end{bmatrix}.$$

Set

$$(B.85) \quad \boldsymbol{\eta}_0^e = [1 \quad a^{-1} \quad 1 \quad 0]^T, \quad \boldsymbol{\eta}_0^o = [x-s \quad 1 \quad 1 \quad -1]^T.$$

These values incorporate the terminal conditions at $m=0$. Direct multiplication in (B.84) reproduces the signed terminal minors above. Hence $\boldsymbol{\eta}_m^\tau = \mathsf{T}_{\text{fold}}^m \boldsymbol{\eta}_0^\tau$ for every $m \geq 0$.

A direct determinant calculation gives

$$(B.86) \quad \begin{aligned} \det(\lambda I_4 - \mathsf{T}_{\text{fold}}) &= \lambda^4 - C_0 \lambda^3 + \mathfrak{b} \lambda^2 - a^2 C_0 \lambda + a^4, \\ \mathfrak{b} &= (1+a^2)x^2 + 2as^2 + 2a^2 - 2a - 2a^3. \end{aligned}$$

Apply Cayley–Hamilton to T_{fold} , multiply by $\mathsf{T}_{\text{fold}}^{m-4} \boldsymbol{\eta}_0^\tau$, and extract the first coordinate. For both $Z_m = \mathcal{Q}_m^e$ and $Z_m = \mathcal{Q}_m^o$, this yields

$$(B.87) \quad Z_m - C_0 Z_{m-1} + \mathfrak{b} Z_{m-2} - a^2 C_0 Z_{m-3} + a^4 Z_{m-4} = 0, \quad m \geq 4.$$

Finally, applying T_{fold} three times to (B.85) gives the boundary residuals

$$\text{Res}_0 = Z_0, \quad \text{Res}_1 = Z_1 - C_0 Z_0, \quad \text{Res}_2 = Z_2 - C_0 Z_1 + \mathfrak{b} Z_0,$$

$$\text{Res}_3 = Z_3 - C_0 Z_2 + \mathfrak{b} Z_1 - a^2 C_0 Z_0,$$

namely

	Res_0	Res_1	Res_2	Res_3
$Z_m = \mathcal{Q}_m^e$	1	$a(1 - 2\zeta_s)$	$a^2(1 - 2\zeta_s)$	a^3
$Z_m = \mathcal{Q}_m^o$	$x - s$	$-[x(1 + a^2) + 2as]$	$a^2(x - s)$	0

Thus, as a formal-power-series identity,

$$\Pi(z) \sum_{m \geq 0} Z_m z^m = \text{Res}_0 + \text{Res}_1 z + \text{Res}_2 z^2 + \text{Res}_3 z^3.$$

For the even sequence the polynomial on the right factors as $(1 + az)(1 - 2a\zeta_s z + a^2 z^2)$; for the odd sequence it is $(x - s) - [x(1 + a^2) + 2as]z + a^2(x - s)z^2$. These factorizations identify the boundary-residual polynomials. Since $\Pi(0) = 1$, division by Π in $\mathbb{R}[[z]]$ gives the two literal formal-power-series identities

$$(B.88) \quad \sum_{m \geq 0} \mathcal{Q}_m^e z^m = (1 + az)(1 - 2a\zeta_s z + a^2 z^2) \Pi(z)^{-1},$$

and

$$(B.89) \quad \sum_{m \geq 0} \mathcal{Q}_m^o z^m = ((x - s) - [x(1 + a^2) + 2as]z + a^2(x - s)z^2) \Pi(z)^{-1}.$$

Appendix C. Divided differences for the folded Chebyshev formulas.

The preceding transfer calculation gives two rational generating functions. Coefficient extraction converts them to the divided-difference formulas used in the half-angle chord parametrization. Fix $0 < a < 1$ and set $U_{-1} = 0$. The Chebyshev generating function gives

$$\sum_{m \geq 0} (u - \zeta_s)(U_m(u) + U_{m-1}(u)) \omega_{\text{gf}}^m = \frac{(u - \zeta_s)(1 + \omega_{\text{gf}})}{1 - 2u\omega_{\text{gf}} + \omega_{\text{gf}}^2},$$

and, with $\mathbf{B}_s^{\text{fold}} = [x(1 + a^2) + 2as]/(2a)$,

$$\sum_{m \geq 0} ((x - s)u - \mathbf{B}_s^{\text{fold}}) U_m(u) \omega_{\text{gf}}^m = \frac{(x - s)u - \mathbf{B}_s^{\text{fold}}}{1 - 2u\omega_{\text{gf}} + \omega_{\text{gf}}^2}.$$

Let $u, v \in \mathbb{R}$ satisfy the two symmetric relations in (5.34). Taking the corresponding divided difference of the Chebyshev generating identities and putting $\omega_{\text{gf}} = az$ recovers (B.88) and (B.89). Coefficient comparison gives (5.35)–(5.36), including the derivative value when $u = v$. The underlying polynomial identities remain valid when $\Delta_{\text{fold}} < 0$, without choosing a square-root branch.

Appendix D. Exhaustion of the endpoint-side chord sheet.

LEMMA D.1 (Endpoint-side chord-sheet exhaustion). *Let $K \geq 3$ and let $j \geq 1$ be an integer. Under the conventions of (5.40) and (5.41), if*

$$\frac{(j+1)\pi}{K} \leq \frac{\pi}{2},$$

then every point of the ordered middle branch in the corresponding open positive gap has a selected endpoint-side j -chord representation with $j\pi/K < \theta < (j+1)\pi/K$, and every θ in this interval gives a point of that branch. The signed root is $(-1)^j s$, its magnitude is $s_{\min}(xI_{K-1} - A_{K-1})$, and the two folded variables are distinct. This includes a gap adjacent to zero. Separately, if K is odd and $j = (K-1)/2$, the same bidirectional statement holds from the positive spectral endpoint to the center of the single central gap, with $j\pi/K < \theta \leq \pi/2$.

Proof. Continue the ordered middle root $s_*(x)$ through the open gap. Write $\Delta_*(x) = \Delta_{\text{fold}}(x, s_*(x))$ for its folded discriminant, let $y(x) \leq z(x)$ be the reconstructed nonnegative half variables on the real sheet issuing from the endpoint germ, and set $\theta_{\text{ch}}(x) = \arccos y(x)$. Let $\mathcal{C}$ be the set of gap points for which

$$\Delta_*(x) > 0, \quad \frac{j\pi}{K} < \theta_{\text{ch}}(x) < \frac{(j+1)\pi}{K}, \quad u_K^{\max} < z(x) < \mathbf{c}_h.$$

On $\mathcal{C}$, the reversible folded determinant identities give $\mathcal{W}_K(y(x)) = \mathcal{W}_K(z(x))$; uniqueness on the endpoint-side outer lobe then gives

$$z(x) = z_K(\mathcal{W}_K(y(x))).$$

Thus $\mathcal{C}$ is precisely the selected endpoint-side chord sheet. At the upper gap endpoint, $z = \mathbf{c}_h$ and $\theta_{\text{ch}} = j\pi/K$. There $z = \mathbf{c}_h > 1 \geq y$, so $\Delta_* > 0$ persists immediately inside the gap. Moreover, Theorem 5.2 gives $\text{sgn } s_* = (-1)^j$; the signed chord equation therefore places θ_{ch} in $(j\pi/K, (j+1)\pi/K)$. Since $|s_*| > 0$, the channel identity gives $z < \mathbf{c}_h$, while continuity keeps $z > u_K^{\max}$. Hence $\mathcal{C}$ is nonempty. The ordered eigenvalues of the continuous real symmetric pencil $B_{K-1}(x)$ are continuous in x , including at an eigenvalue crossing, so the continuation passes through a multiple least singular value. The folded pair $u = 2z^2 - 1$, $v = 2y^2 - 1$ and its reconstructed half variables are continuous, while the sheet inequalities are strict. Thus $\mathcal{C}$ is relatively open in the gap.

It is also relatively closed. At a relative closure point the chord identity and all sheet inequalities persist weakly. The possible closure cases are as follows. (1) If $\Delta_*(x) = 0$, then the reconstructed half variables coincide, which is impossible because

$$z \geq u_K^{\max} > \cos \frac{\pi}{K} \geq \cos \frac{j\pi}{K} \geq y.$$

(2) At an interior point of a positive gap, including one adjacent to zero, both half variables retain their signs: $z > 0$, while $y = 0$ would force $x = 0$, outside the open positive half. (3) Neither the outer variable can reach $\mathbf{c}_h$ nor the inner variable a nodal endpoint: in either case the common level is zero, the endpoint-side solution is $z = \mathbf{c}_h$, and (5.41) gives $s = 0$, impossible inside a spectral gap. (4) The outer variable remains in its endpoint-side component, since its first exit point would be $u_K^{\max}$, contradicting (5.44). Passage through $z = 1$ is harmless; only its coordinate changes. Hence every relative closure point remains in $\mathcal{C}$. Since the open gap is an interval, $\mathcal{C}$ is the entire gap. This proves the first exhaustion direction: every actual gap point lies on the selected endpoint-side chord sheet.

For the converse direction, the reconstructed angle θ_{ch} extends continuously to the closed spectral gap. The exact folded variables at the two endpoints give

$$\theta_{\text{ch}}(\lambda_{K-1,j+1}) = \frac{(j+1)\pi}{K}, \quad \theta_{\text{ch}}(\lambda_{K-1,j}) = \frac{j\pi}{K}.$$

For every $\theta_0 \in (j\pi/K, (j+1)\pi/K)$, the intermediate value theorem therefore supplies an interior gap point x with $\theta_{\text{ch}}(x) = \theta_0$. The classification just proved and uniqueness of the endpoint-side outer solution then give

$$y(x) = \cos \theta_0, \quad z(x) = z_K(\mathcal{W}_K(\cos \theta_0)),$$

so the corresponding abscissa and signed root are exactly those of the prescribed j -chord. Thus every chord parameter produces an actual point of the ordered middle branch. By [Theorem 5.2](#), its positive magnitude is $s_{\min}(xI_{K-1} - A_{K-1})$.

When K is odd, the same argument applies to the positive half of the single central gap, continuing from the positive spectral endpoint to $x = 0$. Because $z > 0$, the identity $x = (2r/\mathfrak{c}_h) \cos \theta z$ forces $\theta = \pi/2$ at $x = 0$. Thus every parameter on that half occurs, and reflection covers the negative half. $\square$

Appendix E. Scalar estimate for the lower central-gap comparison.

This appendix proves the strict scalar inequality (5.59) through finite estimates valid uniformly over the three cases specified below.

Put $b = u_+/u_- = 1/a$, and define

$$Q_b = b^2 - 2b\rho + 1, \quad S_b = b^2 + 2b\omega_L + 1.$$

The Chebyshev identity gives $u_-^2 Q_b = 1$, and

$$(E.90) \quad \mathfrak{D}_L = S_b/Q_b, \quad \Delta_\xi := \rho - \xi = \Delta_\omega Q_b/S_b.$$

Indeed, $u_+ = bu_-$ and $u_-^2 = Q_b^{-1}$, so

$$\mathfrak{D}_L = 1 + 2(\rho + \omega_L)u_-u_+ = 1 + \frac{2b(\rho + \omega_L)}{Q_b} = \frac{S_b}{Q_b},$$

and the formula for Δ_ξ follows from (5.58). The ratio U_L/U_{L-1} is strictly increasing above the largest zero of U_{L-1} , and equals 1 at $\rho_{\text{ref}} = \cos \frac{\pi}{2L+1}$. Hence $\rho > \rho_{\text{ref}}$. Since $S_b - Q_b = 2b(\rho + \omega_L) > 0$,

$$\xi > \rho_{\text{ref}} - \Delta_\omega > 2\omega_L - 1 > 2\omega_L^2 - 1 = \cos \frac{\pi}{L}.$$

Therefore $f_L(x) = (1+x)U_{L-1}(x)$ is positive, strictly increasing, and strictly log-concave on $[\xi, \rho]$. In particular, $\mathfrak{d}_\rho = (\log f_L)'(\rho) > 0$, and

$$\mathfrak{d}_\rho = \begin{cases} \frac{L(b-\rho)-1}{\rho^2-1}, & \rho \neq 1, \\ \frac{2L^2+1}{6}, & \rho = 1 \end{cases}.$$

The endpoint value follows directly from $U_{L-1}(1) = L$ and $U'_{L-1}(1) = L(L^2-1)/3$. Moreover,

$$\log f_L(\rho) - \log f_L(\xi) > \Delta_\xi \mathfrak{d}_\rho, \quad f_L(\xi)^2 < f_L(\rho)^2 e^{-2\Delta_\xi \mathfrak{d}_\rho} < \frac{f_L(\rho)^2}{1 + 2\Delta_\xi \mathfrak{d}_\rho}.$$

Here the first strict inequality is the integral form of strict log-concavity. Since $f_L(\rho)^2 = (1+\rho)^2/Q_b$ and $\Delta_\xi = \Delta_\omega Q_b/S_b$, substitution into (5.59) reduces it to

$$(E.91) \quad \mathcal{N} = (\rho + \omega_L)(S_b + 2\Delta_\omega Q_b \mathfrak{d}_\rho) - (1 + \omega_L)(1 + \rho)^2 > 0.$$

All three cases use

$$(E.92) \quad \Delta_\omega = 1 - \cos \frac{\pi}{2L} < \frac{\pi^2}{8L^2} < \frac{5}{4L^2}, \quad \Delta_\omega L < 1.$$

Suppose first that $\rho > 1$. Put $u = b - \rho$ and $\Delta_+ = \rho^2 - 1$. Since $\mathfrak{d}_\rho > 0$, $u > 1/L$. Keeping ρ fixed, direct differentiation gives

$$\frac{\partial_b \mathcal{N}}{2(\rho + \omega_L)} = \rho + \omega_L + u - \Delta_\omega L + \frac{\Delta_\omega(3Lu^2 - 2u)}{\Delta_+}.$$

The right-hand side is strictly increasing for $u \geq 1/L$ and is positive at $u = 1/L$ because $\Delta_\omega L < 1$. Hence $\mathcal{N}$ exceeds its value at $b = \rho + 1/L$. With $s_0 = \rho + \omega_L$ and $\omega_+ = 1 + \omega_L$, that value is

$$\Upsilon(s_0) = s_0 \left[\left(s_0 + \frac{1}{L} \right)^2 + \omega_+ \Delta_\omega \right] - \omega_+ (s_0 + \Delta_\omega)^2.$$

For $s_0 \geq \omega_+$,

$$\Upsilon''(s_0) = 6s_0 + 4/L - 2\omega_+ > 0, \quad \Upsilon'(\omega_+) = 2\omega_L \omega_+ + 4\omega_+/L + 1/L^2 > 0,$$

and

$$\Upsilon(\omega_+) = \omega_+(-2\Delta_\omega + 2\omega_+/L + 1/L^2) > 0.$$

Thus (E.91) holds when $\rho > 1$. If $\rho = 1$, then $b = (L + 1)/L$, $Q_b = L^{-2}$, and direct substitution, using the endpoint value of $\mathfrak{d}_\rho$, gives

$$\mathcal{N} = \Upsilon(1 + \omega_L) + \frac{2(1 + \omega_L)\Delta_\omega}{L^2} \frac{2L^2 + 1}{6} > 0.$$

The first term is positive by the preceding calculation, and every factor in the correction term is positive.

For $\rho < 1$, set

$$v = 1 - \rho, \quad \Delta_b = b - 1, \quad u = \Delta_b + v, \quad \Delta_- = v(2 - v).$$

Because $b > 1$, we have $0 < v < u$. Moreover, $\rho^2 - 1 < 0$, so the positivity of $\mathfrak{d}_\rho$ in its quotient formula forces $L(b - \rho) - 1 < 0$. Since $u = b - \rho$,

$$(E.93) \quad 0 < v < u < 1/L.$$

With ρ fixed, write $\mathcal{N}_\rho^<(\cdot)$ for $\mathcal{N}$ viewed as a function of $\Delta_b = b - 1$. A direct expansion of (E.91) gives

$$\mathcal{N}_\rho^<(\Delta_b) - \mathcal{N}_\rho^<(0) = \frac{\Delta_b(\rho + \omega_L)}{\Delta_-} \mathfrak{q}(u),$$

where

$$\mathfrak{q}(u) = \Delta_- [2(2 - \Delta_\omega) + u - v] + 2\Delta_\omega [u + v - L(u^2 + uv + v^2) - L\Delta_-].$$

The function $\mathfrak{q}$ is a concave quadratic. Moreover,

$$\mathcal{N}_\rho^<(0) = \frac{v}{2 - v} [(2 - \Delta_\omega)(2 - v)^2 - 2\Delta_\omega^2 - 4\Delta_\omega L(2 - v - \Delta_\omega)].$$

The following elementary bounds hold; the second follows from $\rho > \rho_{\text{ref}}$ and $1 - \rho_{\text{ref}} < \pi^2/[2(2L + 1)^2] < 5/(2L + 1)^2$:

$$(E.94) \quad \Delta_\omega < \frac{5}{4L^2}, \quad v < \frac{5}{(2L + 1)^2} \leq \frac{1}{5}.$$

Although $5/(2L+1)^2 = 1/5$ when $L = 2$, the preceding strict bound still gives $v < 1/5$. These bounds show that the bracketed expression is positive, since it equals

$$(2-v)[(2-\Delta_\omega)(2-v)-4\Delta_\omega L] + 2\Delta_\omega^2(2L-1)$$

and, using $L \geq 2$,

$$(2-\Delta_\omega)(2-v) > \frac{27}{16} \frac{9}{5} = \frac{243}{80} > \frac{5}{2} \geq 4\Delta_\omega L.$$

At the endpoints of (E.93),

$$\mathfrak{q}(v) = v[(2-v)(4-2\Delta_\omega) + 4\Delta_\omega - 4\Delta_\omega L(1+v)] > 0,$$

$$\mathfrak{q}(1/L) = v[(2-v)(4-2\Delta_\omega + 1/L - v) - 4\Delta_\omega L] > 0.$$

At the two endpoints, the positive leading terms exceed

$$\frac{9}{5} \left(4 - \frac{5}{8}\right) = \frac{243}{40}, \quad \frac{9}{5} \left(4 - \frac{5}{8} - \frac{1}{5}\right) = \frac{1143}{200},$$

while the negative terms have magnitudes less than $(5/L)(6/5) \leq 3$ and $5/L \leq 5/2$, respectively. Concavity therefore gives $\mathfrak{q}(u) > 0$, proving (E.91), and hence (5.59), in all three cases.

Appendix F. Signed-root certification in the lower central gap.

This appendix verifies the folded substitution and second-singular-value separation used in the main proof of $d_m > c_{m+1}$.

At $x = x_*$ and $s = \pm\tau_*$, where $\tau_* = c_L$, the real pair $u = \xi$, $v = -\omega_L$ satisfies the two symmetric relations in (5.34). Indeed, its sum and product are $\xi - \omega_L$ and $-\omega_L \xi$, and the required identities follow from

$$\mathfrak{D}_L = \frac{\zeta + \omega_L}{\zeta - \rho}, \quad \xi = \frac{\rho(\zeta + 1) + \zeta(\omega_L - 1)}{\zeta + \omega_L}.$$

Positivity of $\xi + \omega_L$ selects the distinct-variable branch of the divided difference in (5.36). For convenience, write the folded numerator as

$$\mathfrak{n}_s(t) = -[x(\zeta - t) + s(1 + t)]U_{L-1}(t).$$

Put

$$\mathsf{X}_\xi = x_*(\zeta - \xi), \quad \mathsf{Y}_\xi = \tau_*(1 + \xi), \quad \mathsf{X}_\omega = x_*(\zeta + \omega_L), \quad \mathsf{Y}_\omega = \tau_*\Delta_\omega,$$

and set $U_\xi = U_{L-1}(\xi)$, $U_{-\omega} = U_{L-1}(-\omega_L)$. The identities

$$\mathfrak{D}_L = 2a(\zeta + \omega_L)u_+^2, \quad \mathfrak{D}_L(\rho + \omega_L)(\zeta - \xi) = (\zeta + \omega_L)(1 + \xi)$$

give $\mathsf{Y}_\omega \mathsf{Y}_\xi = \mathsf{X}_\omega \mathsf{X}_\xi$. The identity $u_-^2 Q_b = 1$ gives $Q_b > 0$, and $S_b - Q_b = 2b(\rho + \omega_L) > 0$, so $\mathfrak{D}_L = S_b/Q_b > 1$. Moreover, $L \geq 2$ gives $\omega_L > 1/2$, while $\rho > \rho_{\text{ref}} > \omega_L$; hence

$$\rho + \omega_L > 2\omega_L > 1 > 1 - \omega_L > 0.$$

Consequently,

$$\frac{\mathsf{X}_\omega^2}{\mathsf{Y}_\omega^2} = \frac{\mathfrak{D}_L(\rho + \omega_L)}{1 - \omega_L} > 1,$$

so, with $\mathfrak{r}_* = X_\xi/Y_\xi = Y_\omega/X_\omega \in (0, 1)$, the common factor $a^{L-1}/(\xi + \omega_L)$ in (5.36) is positive, and the two numerator differences are

$$\begin{aligned} \mathfrak{n}_{\tau_*}(\xi) - \mathfrak{n}_{\tau_*}(-\omega_L) &= (1 + \mathfrak{r}_*)(X_\omega U_{-\omega} - Y_\xi U_\xi), \\ \mathfrak{n}_{-\tau_*}(\xi) - \mathfrak{n}_{-\tau_*}(-\omega_L) &= (1 - \mathfrak{r}_*)(X_\omega U_{-\omega} + Y_\xi U_\xi). \end{aligned}$$

Their product is positive by (5.59), because

$$U_{-\omega}^2 = \frac{1}{1 - \omega_L^2}, \quad X_\omega^2 U_{-\omega}^2 = \frac{\tau_*^2 \mathfrak{D}_L(\rho + \omega_L)}{1 + \omega_L} > Y_\xi^2 U_\xi^2.$$

Thus

$$(F.95) \quad q_{2L-1}(x_*, \tau_*) q_{2L-1}(x_*, -\tau_*) > 0.$$

At $x = 0$, the second-smallest singular value of A_{2L-1} is

$$s_{L,2}^{(0)} = \sqrt{1 + a^2 - 2a \cos \frac{\pi}{L}}.$$

Indeed, after separating odd and even coordinates, the two rectangular off-diagonal blocks have Gram matrices equal to the $(L-1)$ -dimensional tridiagonal Toeplitz matrix with diagonal $1 + a^2$ and off-diagonal a . Their singular values therefore have squares

$$1 + a^2 + 2a \cos \frac{k\pi}{L}, \quad 1 \leq k \leq L-1,$$

and each occurs twice, while the remaining singular value is zero. We claim

$$(F.96) \quad x_* + \tau_* < 2\sqrt{a} \sin \frac{\pi}{2L} \leq s_{L,2}^{(0)}.$$

Indeed, recall $\rho_{\text{ref}} = \cos \frac{\pi}{2L+1}$. Since $\rho > \rho_{\text{ref}}$ and U_{L-1} is strictly increasing above its largest zero,

$$u_- = U_{L-1}(\rho) > U_{L-1}(\rho_{\text{ref}}) =: u_{-, \text{ref}}, \quad u_{-, \text{ref}}^{-2} = 2(1 - \rho_{\text{ref}}).$$

The identities $\tau_* = a/u_-$ and $\tau_*^2 = 2a(\zeta - \rho)$ imply $\zeta - \rho < a(1 - \rho_{\text{ref}}) < 1 - \rho_{\text{ref}}$. Also

$$\frac{x_*^2}{2a} = \Delta_\omega \frac{\rho + \omega_L}{\zeta + \omega_L} < \Delta_\omega.$$

Put

$$\theta_L = \frac{\pi}{4L}, \quad \Delta_L = \frac{\theta_L}{2L+1}.$$

Then $\Delta_\omega = 2 \sin^2 \theta_L$ and $1 - \rho_{\text{ref}} = 2 \sin^2(\theta_L - \Delta_L)$. Since all quantities are positive, the preceding quotient bound for x_*^2 gives

$$x_* < 2\sqrt{a} \sin \theta_L.$$

Likewise,

$$\tau_*^2 = 2a(\zeta - \rho) < 2a^2(1 - \rho_{\text{ref}}) = 4a^2 \sin^2(\theta_L - \Delta_L),$$

and hence

$$\tau_* < 2a \sin(\theta_L - \Delta_L) < 2\sqrt{a} \sin(\theta_L - \Delta_L),$$

where the last inequality uses $0 < a < 1$, hence $a < \sqrt{a}$. It suffices to prove

$$(F.97) \quad \sin \theta_L + \sin(\theta_L - \Delta_L) < \sin(2\theta_L).$$

The estimates

$$2 \sin \theta_L - \sin(2\theta_L) = 2 \sin \theta_L (1 - \cos \theta_L) < \theta_L^3$$

and

$$\sin \theta_L - \sin(\theta_L - \Delta_L) = \int_{\theta_L - \Delta_L}^{\theta_L} \cos u \, du > \Delta_L \cos \theta_L$$

reduce (F.97) to $\Delta_L \cos \theta_L > \theta_L^3$. Substituting $\Delta_L = \theta_L/(2L+1)$ and $\theta_L = \pi/(4L)$ gives the equivalent condition $2 \cos \theta_L > \theta_L(\pi + 2\theta_L)$. For $0 < \theta_L \leq \pi/8$, using $\pi^2 < 10$,

$$2 \cos \theta_L > 2 - \theta_L^2 \geq \frac{59}{32} > \frac{25}{16} \geq \theta_L(\pi + 2\theta_L).$$

Consequently $x_* + \tau_* < 2\sqrt{a} \sin \frac{\pi}{2L}$. Finally,

$$(s_{L,2}^{(0)})^2 - 4a \sin^2 \frac{\pi}{2L} = (1-a)^2 \geq 0,$$

which proves (5.62).

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